Electrical Power and Energy Systems 155 (2024) 109628

Contents lists available at ScienceDirect

International Journal of Electrical Power and Energy Systems

journal homepage: www.elsevier.com/locate/ijepes

Distributed interactive optimization of integrated electricity-heat energy systems considering hierarchical energy-carbon pricing in carbon markets

Ruijie Liu, Zhejing Bao *, Zhenting Yu, Chenjian Zhang

College of Electrical Engineering, Zhejiang University, Hangzhou 310027, China

ARTICLE INFO

Keywords:
Coordinated scheduling
Energy-carbon pricing
Carbon emissions trading
Integrated energy system
Integrated demand response

ABSTRACT

Considering the impractical direct participation of large number of end-consumers in the carbon emissions trading (CET) market, a distributed interactive optimization framework including the integrated energy supplier and end-consumers is proposed under the ladder-type carbon trading mechanism. A hierarchical Shapley based energy-carbon pricing strategy is developed to reasonably share the energy-carbon costs among end-users, giving incentive to end-users who respond to prices and achieve the participation in CET market. To reduce the costs of carbon emissions and energy consumption, the flexibility of both supplier and end-users are fully explored. Furthermore, the distributed alternating direction method of multipliers (ADMM) method with variable penalty parameters is employed to achieve the coordinated optimization with only limited and necessary information exchanged. Simulation results indicate that the framework proposed can encourage both the source and load to participate in energy-carbon market to reduce energy-carbon costs.

1. Introduction

In recent decades, due to the rampant use of fossil fuels and the consequent surge in greenhouse gas emissions, climate and environmental issues have become one of the most serious challenges people are currently facing. Consequently, various countries and regions have widely recognized the low-carbon economic transformation. For one thing, several low-carbon technologies have been promoted, such as renewable power generations, combined heat and power units, and supercritical and ultra-supercritical generators. And for another, market-based regulatory programs have been launched in many countries to control carbon emissions at low cost.

Many countries and regions have established carbon emissions trading (CET) markets, such as Europe, New Zealand, Australia, China, etc. [1–3]. In general, the government supervision department pre-allocates the free carbon quota to an entity according to its historical carbon emissions or industry baseline. Entities are penalized for exceeding their carbon emissions allotment but can profit if their carbon emissions remain below their quota. Studies mainly focus on the power generation scheduling under CET market, by adding carbon emissions costs to the objective function to realize low-carbon economic operation [4,5]. Recently, the ladder-type carbon trading mechanism (LCTM) has attracted more attention due to its stricter regulation on carbon emissions. The greater the disparity between the carbon emissions and the quota, the higher price of payment/profit will be [6–9]. Since the carbon emissions originate from the energy demand of consumers, end-users should take social responsibility for carbon emission reduction [10]. However, the predominant participants in current market are energy producers and energy-intensive industries such as iron, steel and cement [11]. Direct participation by end-users is impractical due to their large quantity. Literature [10,12,13] has proposed the carbon emission flow (CEF) theory, realizing precise quantification of carbon emissions resulting from each end-user's energy consumption. Based on the CEF theory, reference [14] has established a two-layer low-carbon planning model of the integrated energy system (IES) under LCTM and discussed carbon price setting issues. In [15], a low-carbon optimal learning scheduling based on carbon capture system and CEF theory has been proposed, where the load aggregator responds to the carbon price through demand response (DR). Literature [16] has utilized Shapley values to ascertain the stepped carbon price parameters of LCTM, thereby achieving a low-carbon economic operation of the load side based on CEF theory. A hybrid game optimization has been introduced to derive the optimal trading price not only between micro-grids but also between micro-grid cluster and upper-layer service provide, ultimately reducing carbon emissions through LCTM [8]. Literature [17] has developed a power flow tracing-based method to determine consumers' carbon emission rates, then set the penalty rates for different consumers

* Corresponding author.
E-mail address: zjbao@zju.edu.cn (Z. Bao).

https://doi.org/10.1016/j.ijepes.2023.109628
Received 23 February 2023; Received in revised form 10 October 2023; Accepted 31 October 2023
Available online 9 November 2023
0142-0615/© 2023 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Nomenclature

Indices and Sets

$\mathscr{I}$	set of electric network nodes
$\mathscr{K}$	set of segments of heating network pipelines
$\mathscr{L}$	set of heating network pipelines
$\mathscr{N}$	set of end-users
$\mathscr{O}$	set of branches or pipelines directly connected with a bus or node
$\mathscr{T}$	set of times
a	index of crossing nodes of heating network
i, j, m	index of electric network nodes
if / of	inflow/outflow
k	index of pipeline segments
l	index of heat pipelines
n	index of end-users
o	index of the balanced node
p	index of the heat exchange station node
q	index of heat load nodes
t	time index for electricity network
τ	time index for heating network

Superscript and Subscript

b	bulk power grid
basic	basic load
carbon	carbon emissions
clo	clothing
conv	convertible load
curtail	wind power curtailment
dis	heat dissipation of building
down/up	downward/upward ramping
e	electricity
ex	energy exchange
end	the end of pipeline or time horizon
fc	fence of users' buildings
flex	flexible load
g	coal-fired unit
h	heat
hn	heating network
hs	heat source
in/out	inner/outer layer of pipeline or indoors/outdoors
peak/plain/valley	peak/plain/valley periods
pre	wind power prediction
profit	energy profit rate of the supplier
quote	free carbon quota
r	water flow layer
rad	thermal radiation
s	soil layer
shift	shiftable load
solar	solar radiation
supply	heat power supplied by heating network
system	the whole system including the supplier and users
trade	carbon emissions traded in CET market
u	insulation layer
uncomfy	uncomfortability
w	wind turbine
win	windows of uses' buildings
CHP	CHP unit
I	integrated energy supplier
L	load
P2H	power to heat
PG	power generator
U	end-users

Parameters

Δt	timescale of electricity network [min]
$\Delta\tau$	timescale of heating network [min]
α	solar refraction absorption coefficient
η	energy utilization coefficient
λ	price of the ladder-type carbon trading [$]
ξ	thermal conductivity [W/(m·K)]
σ	interval division of ladder-type carbon trading [tCO_2]
c	carbon emission coefficient [tCO_2/MWh]
d	cost coefficient of users' discomfort [$]
e	energy profit rate of supplier
h	convection heat transfer rate [W/(m^2·K)]
p	cost coefficient [$/MWh]
r	resistance [pu]
x	reactance [pu]
A	heat transfer coefficient of building fence [W/(m^2·°C)]
B	radiative heat flux density [MW/m^2]
D	diameter of pipeline [m]
F	heat capacity of buildings' walls
K	energy conversion efficiency of the CHP unit
M	mass flow rate [kg/s]
I	metabolism rate [met]
R	thermal resistance [(m·K)/W]
S	area [m^2]
Z	buried depth of pipelines [m]

Variables

μ	index of PMV
l	square of current amplitude [pu]
v	square of voltage amplitude [pu]
C	cost [$]
E	carbon emissions [tCO_2]
G	gas power [MW]
H	heat power [MW]
P	electric active power [MW]
Q	electric reactive power [MVar]

through the scenario method. However, a common theme in the aforementioned studies is the presumption of the load side's direct participation in the CET market, which does not align with actual circumstances. Therefore, it is necessary to explore how to incorporate end users into the existing carbon trading market system.

The IES is receiving considerable attention for its potential to enhance energy efficiency and reduce carbon emissions through cascading energy utilization. Given the coupling of various energy types, the traditional electricity market is evolving into a multi-energy trading market. As a result, integrated energy suppliers have become major market participants. They coordinately schedule multi-energy generations, and deliver various energy to end-consumers [18,19]. In order to achieve low-carbon economic operation, the flexibility of the IES needs to be fully exploited. On the supply side, extensive research have been conducted on optimizing the coordination of multi-energy flows [20–25]. On the demand side, the traditional electricity consumers have turned into multi-energy consumers, with integrated demand response (IDR) supplanting the traditional electricity DR. In addition to the traditional means of DR such as load cutting and load shifting, IDR can break the barriers among various energy types owing to multi-energy conversion [19,26,27]. In contrast to conventional electrical power systems which are balanced instantaneously, the balance between heat energy supply and demand is maintained over a period. The heating network and buildings have the inherent potential of thermal energy

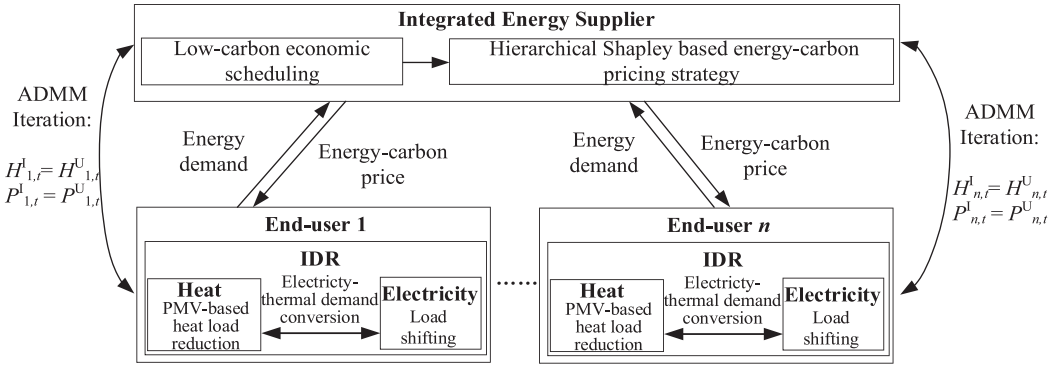


Fig. 1. Coordinated scheduling framework of IES and interaction between the supplier and end-users.

storage. And minor fluctuations in indoor temperature typically go unnoticed by occupants [28–30]. These characteristics of the thermal energy system can provide more flexibility from both the supply and demand sides.

The energy price has been regarded as an effective way to achieve the coordinated operation of multi-participants. A distributed gradient projection iterative algorithm has been proposed to decide the dynamic energy prices and optimal operation strategies of all participants from the perspective of maximizing social welfare [31]. A hierarchical multi-leader multi-follower Stackelberg game model has been proposed to solve the multi-energy trading problem, while proving the uniqueness of Stackelberg equilibrium of multi-energy trading [32]. A Cournot game equilibrium model considering the multi-market strategy interaction among three markets of electricity, gas and heat has been established, in which each IES aims to maximize its total profit in the three markets [33]. As the CET market has evolved, literature on carbon pricing have appeared only recently. With the allocated carbon price signal, each user can be responsible for their own carbon reduction strategy, which is more effective than cooperative incentives. Building on CEF theory, reference [34] has introduced a theory for apportioning carbon emission costs with a fixed carbon price, defined as the carbon emission cost per MWh of energy consumption. Considering the current CET market usually adopts a cap-and-trade policy with ladder-type pricing [1–3,11], the challenge of reasonably and equitably allocating carbon emission costs among end-users remains unsolved, which is the primary focus of this paper. The framework of IES consisting of the supplier and multi-users is structured into outer and inner layers. On the outer layer, supplier takes the system as a whole to participate in the external energy and CET market, and decides the energy-carbon price for end-users by scheduling optimization after receiving energy demand information; on the inner layer, end-users implement IDR according to the prices released by the supplier. Given that carbon emission costs typically serve as punitive measures, unlike profitable energy trading, there's a pressing need for a novel pricing strategy tailored to end-users' carbon emissions.

Based on the literature reviewed above, limited research addresses the integration of IES end-consumers into the CET market. This paper proposes a distributed interactive optimization approach, facilitating indirect participation of end-users in the CET market. The approach encompasses a hierarchical Shapley based energy-carbon pricing strategy, a bi-level distributed coordinated scheduling framework of one supplier and end-users, and an exploration of flexibilities of both the supplier and end-users. By implementing the proposed method, the energy conservation and carbon emission reduction can be achieved with the active engagement of both the supply and demand sides. On the supply side, the integrated energy supplier can reduce carbon emissions by low-carbon economic scheduling of units; on the demand side, end-users can respond to the energy-carbon price provided by the supplier through IDR, thus collaboratively participating in carbon emission reduction. This not only aids in combating global warming but also lessens the likelihood of extreme weather occurrences. The main contributions of this paper are summarized as follows:

- A hierarchical Shapley based energy-carbon pricing strategy is proposed. The strategy seeks a reasonable and fair distribution of energy-carbon costs among end-consumers, considering the LCTM in CET market and the shiftability of flexible load at different periods. The energy-carbon prices are determined by the average marginal effect of end-users' contribution on the collective energy-carbon cost. This not only stimulates end-users' IDR but also incentivizes the establishment of on-site renewable energy sources.
- A bi-level distributed scheduling coordination of the integrated energy supplier and end-users is proposed to achieve the coordinated optimization of multi-participants. Each participant only needs to exchange the necessary information (i.e., demand and price) and carry out its own economic optimal scheduling. Then the global optimal scheduling of all participants can be achieved. Compared with the standard alternating direction method of multipliers (ADMM) method, the convergence speed of ADMM with variable penalty parameters (ADMM-VPP) used in the paper can be improved.
- The flexibilities of both the supplier and users are fully exploited. For the multi-energy supplier, the dynamic model of heating network is established to describe its inherent thermal energy storage characteristics. For end-users, IDR is carried out with the motivation of minimizing the cost of energy-carbon and the induced discomfort. The building's heat storage effect and people's insensitivity to indoor temperature fluctuations are utilized to reduce the costs of carbon emissions and energy consumption.

The remainder of this paper is organized as follows. Section 2 introduces the coordinated scheduling framework of the supplier and end-users. Section 3 proposes the hierarchical Shapley based energy-carbon pricing strategy. In Section 4, the scheduling optimization models of the supplier and end-users are respectively established. The distributed interactive scheduling approach based on ADMM-VPP is presented in Section 5. Case studies of the proposed framework are shown in Section 6. Finally, conclusions and future directions are summarized in Section 7.

2. Coordinated scheduling framework of the integrated energy supplier and multi-consumers

The coordinated scheduling framework of IES and the interaction between the supplier and end-users are described in Fig. 1. As relatively independent individuals, the supplier and end-users are capable of independent decision-making. Only the necessary information, i.e., energy purchasing demand and price, need to be exchanged. In addition to the connection with bulk power grid, the IES has its own energy sources, encompassing the coal-fired unit, wind turbine and CHP unit which

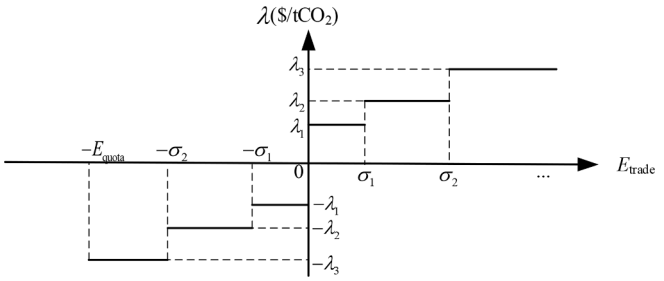


Fig. 2. Ladder-type carbon price.

couples the electricity and thermal networks. At the upper-layer, the integrated energy supplier schedules energy generations and delivers the energy through electricity and thermal networks to meet the multi-energy demands of end-users. At the lower-layer, end-users implement IDR in response to the energy-carbon price released by the supplier, aiming to achieve the minimization of their energy and carbon costs respectively. The coordinated optimization of the integrated energy supplier and multi-end-users is realized by ADMM-VPP. The algorithm effectively safeguards the privacy of individual participants, while achieving the global optimal scheduling of IES. In the distributed optimization, the integrated energy supplier conducts low-carbon economic scheduling of IES considering the constraints of steady power flow in electric distribution network, and dynamic thermal flow in district heat network. Using the hierarchical Shapley algorithm, the energy-carbon prices of end-users are determined by the supplier based on the scheduling results, and passed to corresponding end-users. According to the energy-carbon price, each end-user implements IDR by means of PMV (Predicted Mean Vote) based thermal demand reduction, electricity load shifting and conversion between multi-energy demands, with the goal of minimizing its overall cost. Then end-users derive the energy purchasing strategies and report them to the supplier. Finally, the supplier re-schedules the energy generations and decides the energy-carbon prices, as a response to the updated energy purchasing volume of end-users. The iterative process will be continued until the results converge.

3. Hierarchical Shapley based energy-carbon pricing strategy

3.1. Ladder-type carbon trading mechanism

The integrated energy supplier directly engages in CET market, treating the IES as an integrated entity. The government supervision department pre-allocates the free carbon quota to IES based on its historical carbon emissions or the industry baseline. If the actual carbon emissions are lower than the allocated quota, the surplus can be sold within CET market; otherwise, carbon quota should be purchased for excessive carbon emissions from CET market.

The actual carbon emission of the IES is calculated as:

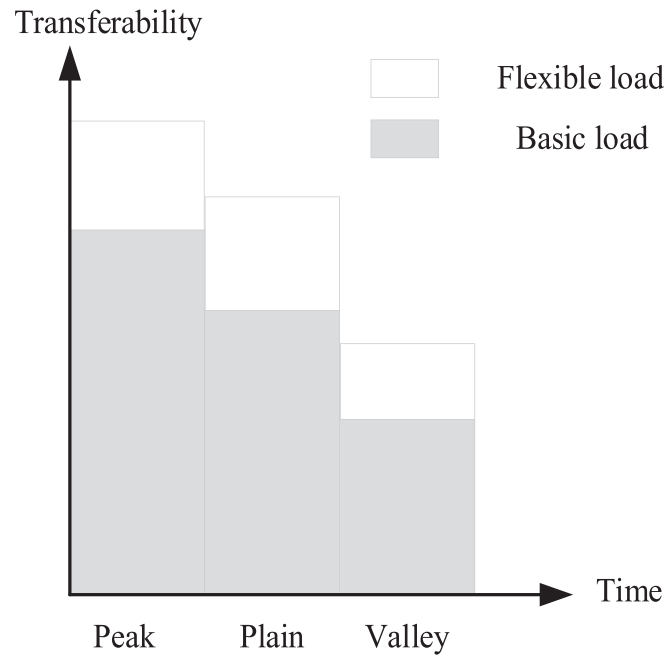


Fig. 4. Load description in two dimensions.

$$E_{\text{system}} = \sum_{t\in\mathcal{T}} \left(c_{\text{g}} P_{\text{g},t} + c_{\text{b}} P_{\text{b},t} + c_{\text{CHP}} G_{\text{CHP},t} + c_{\text{w}} P_{\text{w},t}\right) \Delta t \tag{1}$$

With the allocated free carbon quota E_{quota}, the carbon emission trading volume E_{trade} is given in Eq. (2). The ladder carbon price λ is depicted in Fig. 2, which exhibits a stepwise increase as E_{trade} reaches a new segmentation point σ. The parameters σ and λ are pre-decided by the administration. The total cost or revenue $C^{\text{I}}_{\text{carbon}}$ is formulated in Eq. (3) and depicted in Fig. 3. The value of $C^{\text{I}}_{\text{carbon}}(E^{*}_{\text{trade}})$ is obtained by integrating the variable E_{trade} with respect to $\lambda = \lambda(E_{\text{trade}})$ from 0 to E^{*}_{trade} when $E_{\text{trade}} > 0$, and from E^{*}_{trade} to 0 when $E_{\text{trade}} < 0$.

$$E_{\text{trade}} = E_{\text{system}} - E_{\text{quota}} \tag{2}$$

$$C^{\text{I}}_{\text{carbon}}(E_{\text{trade}}) = \begin{cases} \lambda_3(E_{\text{trade}} + \sigma_2) + \lambda_2(-\sigma_2 + \sigma_1) - \lambda_1\sigma_1 & E_{\text{trade}} \leqslant -\sigma_2 \\ \lambda_2(E_{\text{trade}} + \sigma_1) - \lambda_1\sigma_1 & -\sigma_2 < E_{\text{trade}} \leqslant -\sigma_1 \\ \lambda_1 E_{\text{trade}} & -\sigma_1 < E_{\text{trade}} \leqslant \sigma_1 \\ \lambda_2(E_{\text{trade}} - \sigma_1) + \lambda_1\sigma_1 & \sigma_1 < E_{\text{trade}} \leqslant \sigma_2 \\ \lambda_3(E_{\text{trade}} - \sigma_2) + \lambda_2(\sigma_2 - \sigma_1) + \lambda_1\sigma_1 & E_{\text{trade}} > \sigma_2 \end{cases} \tag{3}$$

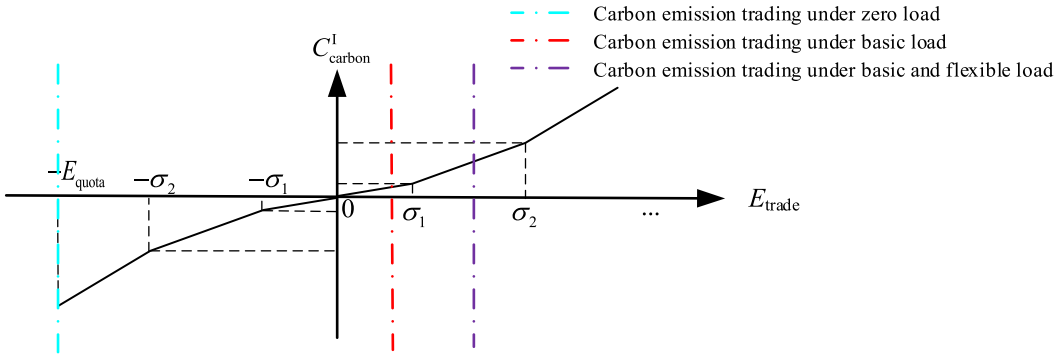


Fig. 3. Carbon emission cost/revenue under LCTM.

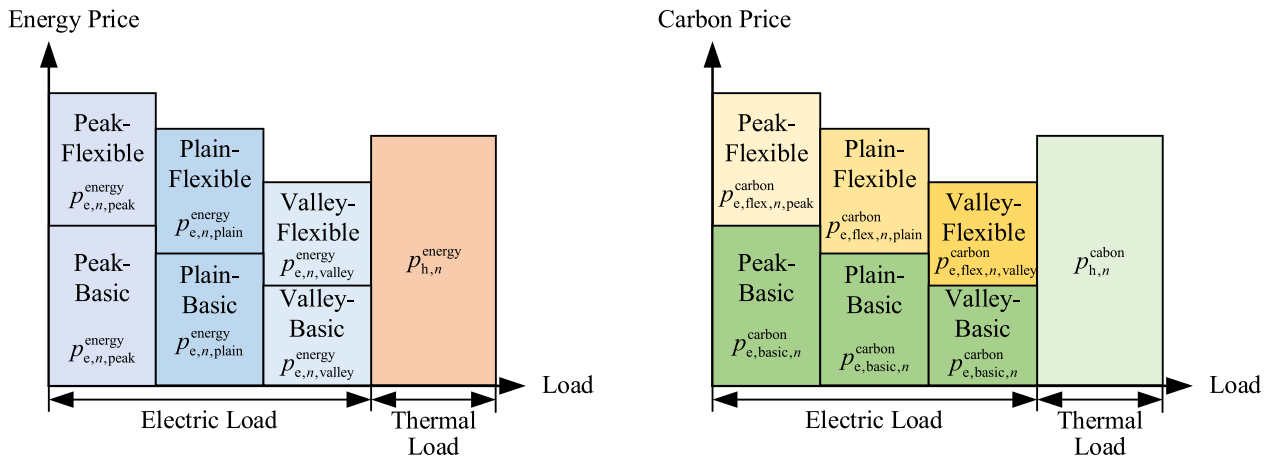


Fig. 5. Energy and carbon pricing strategy of the integrated energy supplier.

3.2. Hierarchical Shapley based energy-carbon pricing strategy under LCTM

Since the carbon emissions actually stem from the energy demand of consumers, end-users should be regarded as the primary cause of carbon emissions. Establishing a pricing mechanism that compels energy consumers to bear associated costs and responsibilities is crucial, yet it presents challenges due to the intricate and competitive dynamics among end-users. The energy-carbon pricing mechanism can motivate end-users to implement IDR, achieving energy conservation and emission reduction.

Given the directive nature of energy-carbon pricing in guiding users towards IDR, it's essential to classify loads in different dimensions. As Fig. 4 shows, users' load can be described from two dimensions, i.e., transferability (basic/flexible) and time (peak/plain/valley). Within the transferability dimension, there exists two kinds of load, basic and flexible. The former is essential for the operation of end-users, which is unshiftable; while the latter can be shifted in time dimension or converted among energy types. The characteristics in the two dimensions have different contributions. For example, the energy price during peak periods is higher than that during valley periods, and the carbon price of basic load is different from that of flexible load as a result of LCTM. Therefore, in order to accurately apportion the cost of energy and carbon emission among end-users, a hierarchical Shapley based approach is proposed to determine the energy-carbon price of each user, making use of the additivity characteristics of Shapley algorithm.

The Shapley value is deemed helpful when quantifying the allocation problem among multi-members. It can reflect the interaction among consumers and quantify the contribution of each consumer to the overall costs of energy consumption and carbon emissions. For the basic load which does not participate in IDR, differentiation across time periods is unnecessary. The Shapley method is used to calculate its corresponding energy-carbon cost allocation for the consideration of simplifying computation. For the flexible load, further consideration across different time periods is essential. Aumann-Shapley value method, referred to as A-S hereinafter for simplicity, is adopted due to its superior computational speed when dealing with larger user datasets. The pricing strategy of the integrated energy supplier is given in Fig. 5. Considering the flexibility of thermal loads, the energy-carbon price for thermal demand adopts a uniform one. In addition, following the usual practice, energy prices do not differentiate between the basic load and the flexible load. Different prices are distinguished by different colors in Fig. 5.

The hierarchical Shapley based pricing strategy is detailed as follows. During the following calculation, since CHP unit operates as a cogenerator, its energy-carbon cost is allocated to electricity and heat generation based on the heat-to-power ratio. Moreover, in order to simplify calculation, the heat load is categorized as virtual 'basic'/'flexible' and 'peak'/'plain'/'valley' in same proportion of the electric load in the following first and second steps, and is merged in the last step.

First, the initial carbon emission cost or benefit is allocated to end-users. When the coalition is empty (i.e., zero load), the carbon emissions E_{system} is zero, and E_{trade} equals to $-E_{\text{quota}}$. It can be seen from Fig. 3 that there exists carbon emission benefit for the IES under LCTM. The benefit is allocated to each end-user reasonably based on the proportion of each user's load, given in Eqs. (4)–(5).

$$C^{\text{I}}_{\text{carbon,init}} = C^{\text{I}}_{\text{carbon}}(-E_{\text{quota}}) \tag{4}$$

$$C^{\text{U}}_{\text{carbon,init},n} = \frac{\sum_{t\in\mathcal{T}}\left(P^{\text{supply,init}}_{n,t} + H^{\text{hn,init}}_{n,t}\right)C^{\text{I}}_{\text{carbon,init}}}{\sum_{t\in\mathcal{T}}\sum_{n\in\mathcal{N}}\left(P^{\text{supply,init}}_{n,t} + H^{\text{hn,init}}_{n,t}\right)} \tag{5}$$

Second, for the basic load, the Shapley method is used to calculate the energy-carbon cost allocation of each user. In the paper, end-users' basic load is the member of Shapley method. By calculating all potential coalitions and averaging the marginal effects of member i when joining coalitions, Shapley can derive the cost allocation x_i apportioned to member i as:

$$x_i = \sum_{S\subseteq N/i} prob(S)(c(S\cup\{i\}) - c(S)) \tag{6}$$

where S is a coalition without member i. $S\cup\{i\}$ represents a new coalition when member i joins S. $c(S\cup\{i\})-c(S)$ indicates the marginal effect of member i joining the coalition S. When calculating the energy price, $c(S)$ stands for the energy cost caused by S. Similarly, when calculating the carbon price, $c(S)$ represents the LCTM cost. $prob(S)$ denotes the occurrence probability of coalition S, and can be calculated as:

$$prob(S) = \frac{n_S!(n_N - n_S - 1)!}{n_N!} \tag{7}$$

where n_N is the number of all members. n_S is the number of members within coalition S.

The Shapley method can guarantee the fairness of users. Firstly, users with identical marginal contributions to the overall energy and carbon costs will share the same energy-carbon price. Secondly, the property of independence is also satisfied, implying that the energy-carbon price of users remains unaffected by the order in which users are selected. By separately calculating the marginal effect of user n's basic load on the energy consumption cost and carbon emission cost of the system, the energy price and carbon price can be obtained.

Next, for the flexible load, based on the pricing results of basic load, the A-S value method is applied to apportion the cost of each user across different time periods. This is mathematically expressed as:

$$x_i = \int_0^1 \frac{\partial c(t\boldsymbol{P})}{\partial P_i} dt \tag{8}$$

where $\boldsymbol{P}$ is the vector composed of users' flexible load at different time periods $P_{1,\text{flex,peak}}, P_{1,\text{flex,plain}}, P_{1,\text{flex,valley}}, \cdots, P_{n_N,\text{flex,valley}}$, which is uniformly denoted as $P_1, P_2, \cdots, P_{n_N}$.

The integral expression in Eq. (8) can be numerically calculated by discretization. Dividing the load P_i into N small segments equally, the allocation of load i is expressed as:

$$x_i = \sum_{k=0}^{N-1} \left[c\left(\frac{k\boldsymbol{P}}{N} + \left[\underbrace{\frac{P_1}{N}, \cdots, \frac{P_i}{N}}_{i}, 0, \cdots, 0 \right] \right) - c\left(\frac{k\boldsymbol{P}}{N} + \left[\underbrace{\frac{P_1}{N}, \cdots, \frac{P_{i-1}}{N}}_{i-1}, 0, \cdots, 0 \right] \right) \right] \tag{9}$$

Compared to the computational complexity 2^{n_N} of the Shapley method, that of A-S is significantly reduced to $N \times n_N$ when n_N is larger. The choice of N usually considers the balance between the computational accuracy and speed [35]. When N is set to 9, the result differences caused by swapping user calculation orders are only 0.34 % through practical verification. Therefore, A-S method can be regarded as a fair algorithm.

Finally, based on the aforementioned allocation results, the hierarchical energy and carbon prices are calculated respectively. Considering the profit rate e_{profit} set by the supplier, the final energy price is derived according to Eqs. (10)–(11). For the carbon price, since it is a punitive or incentive measure established by the government, the carbon cost or benefit is shared reasonably among the users without any profit accruing to the supplier. This is shown in Eqs. (12)–(13).

In Fig. 3, the average slope between the red and blue dash-dotted lines determines the carbon price of the basic load. Upon adding the flexible load, the carbon cost difference between the red and purple lines represents the additional carbon cost attributable to the flexible load. Consequently, the carbon price of basic load is different from that of flexible load due to the LCTM. The proposed hierarchical pricing strategy reflects the differential pricing of carbon emissions trading inherent in CET market.

The energy-carbon pricing strategy serves as the connection between the integrated energy supplier and end-users. With the interaction of pricing strategy and scheduling results, the global coordination of the IES can be achieved. While there is no direct energy trading among end-users, their individual energy purchasing strategies can exert mutual influence.

4. Scheduling optimization models of the integrated energy supplier and end-users

4.1. Integrated energy supplier

4.1.1. Optimization objective

The optimization objective of the integrated energy supplier is to minimize the total costs during the scheduling time horizon, including the carbon emissions costs and energy generation costs:

$$\min C^{\text{I}} = C^{\text{I}}_{\text{energy}} + C^{\text{I}}_{\text{carbon}}$$
$$C^{\text{I}}_{\text{energy}} = \sum_{t\in\mathcal{T}} \left(p_{\text{b},t} P_{\text{b},t} + p_{\text{CHP}} G_{\text{CHP},t} + p_{\text{g}} P_{\text{g},t} + p_{\text{curtail}} \left(P^{\text{pre}}_{\text{wind}} - P_{\text{wind}} \right) \right) \Delta t \tag{14}$$

The detailed calculation of C^{I}_{carbon} is given in Eq. (3). Besides, in order to promote wind power consumption, wind curtailment penalty is included.

4.1.2. Constraints

$$p^{\text{energy}}_{\text{e},n,\text{peak/plain/valley}} = e_{profit} \frac{x^{\text{energy}}_{\text{e,flex},n,\text{peak/plain/valley}} + x^{\text{energy}}_{\text{e,basic},n} \sum_{t\in\mathcal{T}_{\text{peak/plain/valley}}} P^{\text{supply}}_{n,t} \Big/ \sum_{t\in\mathcal{T}} P^{\text{supply}}_{n,t}}{\sum_{t\in\mathcal{T}_{\text{peak}}} P^{\text{supply}}_{n,t}} \tag{10}$$

$$p^{\text{energy}}_{h,n} = e_{\text{profit}} \frac{\left(\sum_{t\in\mathcal{T}} x^{\text{energy}}_{\text{h,basic},n} + \sum_{t\in\mathcal{T}_{\text{peak}}} x^{\text{energy}}_{\text{h,flex},n,\text{peak}} + \sum_{t\in\mathcal{T}_{\text{plain}}} x^{\text{energy}}_{\text{h,flex},n,\text{plain}} + \sum_{t\in\mathcal{T}_{\text{valley}}} x^{\text{energy}}_{\text{h,flex},n,\text{valley}} \right)}{\sum_{t\in\mathcal{T}} H^{\text{hn}}_{n,t}} \tag{11}$$

(1) Electricity network

Since the topology of electricity distribution network in IES is radial, the Distflow model after second-order cone relaxation is introduced to establish the AC power flow [36]:

$$\begin{cases} p^{\text{carbon}}_{\text{e,basic},n} = \dfrac{x^{\text{carbon}}_{\text{e,basic},n}}{\sum_{t\in\mathcal{T}} P^{\text{basic}}_{n,t}} \\ p^{\text{carbon}}_{\text{e,flex},n,\text{peak/plain/valley}} = \dfrac{x^{\text{carbon}}_{\text{e,flex},n,\text{peak/plain/valley}}}{\sum_{t\in\mathcal{T}_{\text{peak/plain/valley}}} P^{\text{flex}}_{n,t}} \end{cases} \tag{12}$$

$$p^{\text{carbon}}_{\text{h},n} = \frac{\left(\sum_{t\in\mathcal{T}} x^{\text{carbon}}_{\text{h,basic},n} + \sum_{t\in\mathcal{T}_{\text{peak}}} x^{\text{carbon}}_{\text{h,flex},n,\text{peak}} + \sum_{t\in\mathcal{T}_{\text{plain}}} x^{\text{carbon}}_{\text{h,flex},n,\text{plain}} + \sum_{t\in\mathcal{T}_{\text{valley}}} x^{\text{carbon}}_{\text{h,flex},n,\text{valley}} \right)}{\sum_{t\in\mathcal{T}} H^{\text{hn}}_{n,t}} \tag{13}$$

$$\begin{aligned} P_{\mathrm{PG},i,t} - P_{\mathrm{L},i,t} &= \sum_{(i,j)\in\mathscr{J}} P_{ij,t} - \sum_{(m,i)\in\mathscr{J}} \left(P_{mi,t} - r_{mi} l_{mi,t}\right) \\ Q_{\mathrm{PG},i,t} - Q_{\mathrm{L},i,t} &= \sum_{(i,j)\in\mathscr{J}} Q_{ij,t} - \sum_{(m,i)\in\mathscr{J}} \left(Q_{mi,t} - x_{mi} l_{mi,t}\right) \\ v_{j,t} &= v_{i,t} - 2\left(r_{ij}P_{ij,t} + x_{ij}Q_{ij,t}\right) + \left(r_{ij}^2 + x_{ij}^2\right) l_{ij,t} \\ l_{ij,t} &\geqslant \frac{P_{ij,t}^2 + Q_{ij,t}^2}{v_{i,t}} \end{aligned}, t \in \mathscr{T} \tag{15}$$

$$l_{ij,t} \leqslant l_{\max}, (i,j) \in \mathscr{J}, t \in \mathscr{T} \tag{16}$$

$$v_{\min} \leqslant v_{i,t} \leqslant v_{\max}, i \neq o, i \in \mathscr{J}, t \in \mathscr{T} \tag{17}$$

$$v_{o,t} = 1, t \in \mathscr{T} \tag{18}$$

where P_{ij} and Q_{ij} are the active and reactive powers flowing from node i to j.

The electricity generations of the coal-fired unit are limited by the ramping rate:

$$P_{\mathrm{g}}^{\mathrm{down}} \leqslant P_{\mathrm{g},t+1} - P_{\mathrm{g},t} \leqslant P_{\mathrm{g}}^{\mathrm{up}}, t \in \mathscr{T} \tag{19}$$

The electricity generations are also constrained by their upper and lower bounds:

$$\begin{aligned} P_{\mathrm{g}}^{\min} \leqslant P_{\mathrm{g},t} \leqslant P_{\mathrm{g}}^{\max}, t \in \mathscr{T} \\ P_{\mathrm{w}}^{\min} \leqslant P_{\mathrm{w},t} \leqslant P_{\mathrm{w}}^{\mathrm{pre}}, t \in \mathscr{T} \end{aligned} \tag{20}$$

(2) Heating network [37].

Due to the slow heating transmission, heat energy supply and demand are not balanced in real time. Detailed modeling of the heating network is essential. The most commonly used strategy CF-VT (constant flow and variable supply temperature) in north China is adopted [22,38]. Considering the heating network pipeline with insulation layer, a micro-element-based transient heating network model is established. The pipeline is divided equally into small segments Δx. For each segment, heat energy in the hot water is dissipated first to insulation and subsequently to the surrounding soil in radial direction. Simultaneously, it transfers downwards via water flows in the axial direction. The timescale Δt of the electric power scheduling is not suited for capturing the nuances of the transient heating network. Consequently, a finer timescale $\Delta\tau$ is used for modeling heating network.

The respective thermal resistances between the hot water and the insulation layer, as well as between the insulation and soil layers, can be described as:

$$R_{\mathrm{ru}} = \frac{1}{hD_{\mathrm{in}}} + \frac{1}{2\xi_{\mathrm{u}}} \ln\left(\frac{D_{\mathrm{out}}}{D_{\mathrm{in}}}\right) \tag{21}$$

$$R_{\mathrm{us}} = \frac{1}{2\xi_{\mathrm{s}}} \ln\left[\frac{2Z}{D_{\mathrm{out}}} + \sqrt{\left(\frac{2Z}{D_{\mathrm{out}}}\right)^2 - 1}\right] \tag{22}$$

The heat dissipation of the insulation layer is:

$$C_{\mathrm{u}} \frac{T_{\mathrm{u},l,k,\tau+1} - T_{\mathrm{u},l,k,\tau}}{\Delta\tau} = \frac{\pi\Delta x}{R_{\mathrm{ru}}}\left(T_{\mathrm{r},l,k,\tau} - T_{\mathrm{u},l,k,\tau}\right) - \frac{\pi\Delta x}{R_{\mathrm{us}}}\left(T_{\mathrm{u},l,k,\tau} - T_{\mathrm{s}}\right), \tau \in \mathscr{T}, k \in \mathscr{K}, l \in \mathscr{L} \tag{23}$$

where $C_{\mathrm{u}} = \frac{\pi}{4} c_{\mathrm{u}} (D_{\mathrm{out}}^2 - D_{\mathrm{in}}^2) \Delta x \rho_{\mathrm{u}}$.

Besides heat energy dissipation, heat energy is transferred to the adjacent next segment at the same time, which is modeled as:

$$T_{\mathrm{r},l,k+1,\tau+1} = \frac{\frac{\pi}{4} D_{\mathrm{in}}^2 \rho_{\mathrm{r}} c_{\mathrm{r}} \Delta x T_{\mathrm{r},l,k+1,\tau} + M_l c_{\mathrm{r}} \Delta\tau T_{\mathrm{r},l,k,\tau+1} + \frac{\pi}{R_{\mathrm{ru}}} \Delta x \Delta\tau T_{\mathrm{u},l,k+1,\tau+1}}{\frac{\pi}{4} D_{\mathrm{in}}^2 \rho_{\mathrm{r}} c_{\mathrm{r}} \Delta x + M_l c_{\mathrm{r}} \Delta\tau + \frac{\pi}{R_{\mathrm{ru}}} \Delta x \Delta\tau}$$
$$\tau \in \mathscr{T}, k \in \mathscr{K}, l \in \mathscr{L} \tag{24}$$

At the crossing node a where water flows converge, the mass flow balance is maintained:

$$\sum_{if \in \mathscr{C}_{a,\mathrm{pipe}+}} M_{if} = \sum_{of \in \mathscr{C}_{a,\mathrm{pipe}-}} M_{of} \tag{25}$$

where $\mathscr{C}_{a,\mathrm{pipe}+}$ and $\mathscr{C}_{a,\mathrm{pipe}-}$ are the sets of pipelines that flow into and out of node a.

The water temperature after mixing is:

$$T_{\mathrm{r},of,1,\tau} = \frac{\sum_{if \in \mathscr{C}_{k,\mathrm{pipe}+}} M_{if} T_{\mathrm{r},if,k_{\mathrm{end}},\tau}}{\sum_{if \in \mathscr{C}_{k,\mathrm{pipe}+}} M_{if}}, of \in \mathscr{C}_{a,\mathrm{pipe}-} \tag{26}$$

where k_{end} denotes the last segment of the pipeline.

Assuming that the heat exchange station is located at node p, the heat exchange is formulated as:

$$\left(\sum_{of \in \mathscr{C}_{p,\mathrm{pipe}-}} T_{\mathrm{r},of,1,\tau+1} M_{of} - \sum_{if \in \mathscr{C}_{p,\mathrm{pipe}+}} T_{\mathrm{r},if,k_{\mathrm{end}},\tau} M_{if}\right) c_r = \eta_{\mathrm{ex}} H_{\mathrm{CHP},\tau}, \tau \in \mathscr{T} \tag{27}$$

Similarly, the heat exchange at the heat load node q is:

$$\left(\sum_{if \in \mathscr{C}_{q,\mathrm{pipe}+}} T_{\mathrm{r},if,k_{\mathrm{end}},\tau} M_{if} - \sum_{of \in \mathscr{C}_{q,\mathrm{pipe}-}} T_{\mathrm{r},of,1,\tau+1} M_{of}\right) c_{\mathrm{r}} = \frac{H_{q,\tau}}{\eta_{\mathrm{load}}}, \tau \in \mathscr{T} \tag{28}$$

The water temperature in pipelines is limited by the upper and lower bounds:

$$T_{\mathrm{hn}}^{\min} \leqslant T_{\mathrm{r},l,k,\tau} \leqslant T_{\mathrm{hn}}^{\max}, \tau \in \mathscr{T}, k \in \mathscr{K}, l \in \mathscr{L} \tag{29}$$

The difference of end-users' inlet water temperatures between the beginning and end of the scheduling period is typically kept within a certain range:

$$\Delta T_{\mathrm{hn}}^{\min} \leqslant T_{\mathrm{r},if,k_{end},\tau_{\mathrm{end}}} - T_{\mathrm{r},if,k_{\mathrm{end}},1} \leqslant \Delta T_{\mathrm{hn}}^{\max}, if \in \mathscr{C}_{q,\mathrm{pipe}+} \tag{30}$$

(3) CHP unit

The electricity and heating networks are coupled by CHP units. Back-pressure turbine is used in the paper. Its heat-to-electricity ratio is usually constant, and its operational characteristics is described as:

$$P_{\mathrm{CHP},t} = K_{\mathrm{e}} G_{\mathrm{CHP},t}, t \in \mathscr{T} \tag{31}$$

$$H_{\mathrm{CHP},\tau} = K_{\mathrm{h}} G_{\mathrm{CHP},t}, \tau \in t, t \in \mathscr{T} \tag{32}$$

As shown in Eqs. (31)–(32), the CHP unit integrates the dynamic thermal flow model of heating network with the steady AC power flow model of electricity network, accommodating the distinct timescales t and τ.

The power ramping rate constraint of the CHP unit is:

$$P_{\mathrm{CHP}}^{\mathrm{down}} \leqslant P_{\mathrm{CHP},t+1} - P_{\mathrm{CHP},t} \leqslant P_{\mathrm{CHP}}^{\mathrm{up}}, t \in \mathscr{T} \tag{33}$$

Output power of the CHP unit is bounded by its physical limits:

$$P_{\mathrm{CHP}}^{\min} \leqslant P_{\mathrm{CHP},t} \leqslant P_{\mathrm{CHP}}^{\max}, t \in \mathscr{T} \tag{34}$$

4.2. End-users

4.2.1. Integrated demand responds

End-users can respond to the energy-carbon price released by the integrated energy supplier through IDR. Not only the electricity load can

be shifted from the peak periods to valley periods, but also the heat demand can be converted into electricity demand when electricity energy is cheaper. For example, end-users can either use the heat energy provided by heat supply network or opt for electric air conditioning. Moreover, in contrast to electricity power, real-time balancing of supply and demand isn't imperative for heat power. Buildings inherently offer thermal energy storage capabilities. Additionally, human perception of thermal changes is somewhat ambiguous. Minor fluctuations in heat energy supply typically don't compromise human comfort. Consequently, performing IDR is vital to utilize the flexibility of IES while maintaining end-users' satisfaction.

The electrical load of user n composes two categories, i.e., basic and flexible. And the flexible category further encompasses shiftable and convertible loads. The total electric load should be balanced by the power supply:

$$P_{n,t}^{\text{supply}} = P_{n,t}^{\text{basic}} + P_{n,t}^{\text{flex}} = P_{n,t}^{\text{basic}} + P_{n,t}^{\text{shift}} + P_{n,t}^{\text{conv}} \tag{35}$$

Since the proposed model captures the inherent inertia of heat supply network and buildings, the shifting of heat load in time dimension is not explicitly considered. The heat demand of user n can be satisfied by the heat power both delivered by heating network and generated by the convertible electric power $P_{n,t}^{\text{conv}}$:

$$H_{n,t}^{\text{supply}} = \eta_{\text{P2H}} P_{n,t}^{\text{conv}} + H_{n,\tau}^{\text{hn}}, \tau \in t \tag{36}$$

4.2.2. Optimization objective

The optimization objective of end-user n is to minimize both the energy-carbon cost and the discomfort induced by IDR:

$$\min C_n^{\text{U}} = C_{\text{energy},n}^{\text{U}} + C_{\text{carbon},n}^{\text{U}} + C_{\text{uncomfy},n}^{\text{U}} \tag{37}$$

$$C_{\text{energy},n}^{\text{U}} = \sum_{t\in\mathscr{T}} (p_{\text{e},n,t}^{\text{energy}} P_{n,t}^{\text{supply}} + p_{\text{h},n}^{\text{energy}} H_{n,t}^{\text{hn}}) \tag{38}$$

$$C_{\text{carbon},n}^{\text{U}} = \sum_{t\in\mathscr{T}} (p_{\text{e,basic},n}^{\text{carbon}} P_{n,t}^{\text{basic}} + p_{\text{e,flex},n,t}^{\text{carbon}} P_{n,t}^{\text{flex}} + p_{\text{h},n}^{\text{cabon}} H_{n,t}^{\text{hn}}) - C_{\text{carbon,init},n}^{\text{U}} \tag{39}$$

$$C_{\text{uncomfy},n}^{\text{U}} = d_{\text{shift}} \frac{\left|\Delta P_{n,t}^{\text{shift}}\right|}{P_{n,t}^{\text{shift,init}}} + d_{\text{conv}} \frac{\left|\Delta P_{n,t}^{\text{conv}}\right|}{P_{n,t}^{\text{conv,init}}} \tag{40}$$

where $C_{\text{carbon,init},n}^{\text{U}}$ is derived from Eq. (5). $\left|\Delta P_{n,t}^{\text{shift}}\right|$ and $\left|\Delta P_{n,t}^{\text{conv}}\right|$ are further linearized as:

$$\begin{aligned} &\Delta P_{n,t}^{\text{shift}} - P_{n,t}^{\text{shift},+} + P_{n,t}^{\text{shift},-} = 0 \\ &\Delta P_{n,t}^{\text{conv}} - P_{n,t}^{\text{conv},+} + P_{n,t}^{\text{conv},-} = 0 \\ &P_{n,t}^{\text{shift},+}, P_{n,t}^{\text{shift},-}, P_{n,t}^{\text{conv},+}, P_{n,t}^{\text{conv},-} \geqslant 0 \end{aligned} \tag{41}$$

$$C_{\text{uncomfy},n}^{\text{U}} = d_{\text{shift}} \frac{P_{n,t}^{\text{shift},+} + P_{n,t}^{\text{shift},-}}{P_{n,t}^{\text{shift,init}}} + d_{\text{conv}} \frac{P_{n,t}^{\text{conv},+} + P_{n,t}^{\text{conv},-}}{P_{n,t}^{\text{conv,init}}} \tag{42}$$

4.2.3. Constraints

The shiftable electricity load is modelled as:

$$\begin{cases} P_{n,t}^{\text{shift}} = P_{n,t}^{\text{shift,init}} + \Delta P_{n,t}^{\text{shift}} \\ \sum_{t\in\mathscr{T}} \Delta P_{n,t}^{\text{shift}} = 0 \\ P_{n,t}^{\text{shift}} \geqslant 0 \end{cases} \tag{43}$$

The convertible electric and heat load is described as:

$$\begin{cases} P_{n,t}^{\text{conv}} = P_{n,t}^{\text{conv,init}} + \Delta P_{n,t}^{\text{conv}} \\ H_{n,t}^{\text{conv}} = H_{n,t}^{\text{conv,init}} + \Delta H_{n,t}^{\text{conv}} \\ \sum_{t\in\mathscr{T}} (\Delta H_{n,t}^{\text{conv}} + \eta_{\text{P2H}} \Delta P_{n,t}^{\text{conv}}) = 0 \\ P_{n,t}^{\text{conv}} \geqslant 0 \\ H_{n,t}^{\text{conv}} \geqslant 0 \end{cases} \tag{44}$$

The first-order linear differential equation is used to model users' buildings. The heat dissipation from indoor to outdoor can be formulated as:

$$H_{n,t}^{\text{dis}} = A_n^{\text{fc}} S_n^{\text{fc}} (T_{n,t}^{\text{in}} - T_{n,t}^{\text{out}}) \tag{45}$$

where the heat transfer coefficient A_n^{fc} is influenced by the structure of the fence, such as window and wall. A thicker building envelope generally results in superior insulation.

The heat absorption energy of buildings from solar radiation is:

$$H_{n,t}^{\text{rad}} = \alpha_n S_n^{\text{win}} B_{n,t}^{\text{solar}} \tag{46}$$

Considering the gradual dynamics of heat dissipation, heat absorption energy and indoor heat source, the indoor temperature is expressed by:

$$T_{h,t+1}^{\text{in}} = T_{h,t}^{\text{in}} + \frac{(H_{n,t}^{\text{supply}} + H_{n,t}^{\text{hs}} + H_{n,t}^{\text{rad}} - H_{n,t}^{\text{dis}})\Delta t}{F_n} \tag{47}$$

The PMV index is introduced for evaluating users' indoor thermal comfort. Its simplified expression is [39]:

$$\mu_{\text{PMV}} = 2.43 - \frac{3.76(T_{\text{comfy}} - T_{\text{in}})}{I(R_{\text{clo}} + 0.1)} \tag{48}$$

where I, T_{comfy} and R_{clo} are the thermodynamic parameters. μ_{PMV} is a function of T_{in}.

According to ISO Standard 7730, slight temperature variations are typically imperceptible to users. According to GB/T Standard 18,049 in China, PMV should be limited within [−1, 1] to satisfy users' thermal preferences. Since users tend to be more thermally sensitive in daytime, the PMV index is restricted to [−0.5, 0.5] in daytime, and relaxed to the range [−1, 1] at night [40]:

$$\mu_{\text{PMV},t}^{\min} \leqslant \mu_{\text{PMV}} \leqslant \mu_{\text{PMV},t}^{\max}, t \in \mathscr{T} \tag{49}$$

4.3. Summary of scheduling optimization problems

According to Section 4.1, the scheduling optimization problem of the integrated energy supplier can be formulated in Eq. (50):

$$\begin{cases} \min C^{\text{I}} \\ \text{s.t.} (15) \text{ - } (34) \end{cases} \tag{50}$$

The optimization variables include: (1) the current amplitudes, electric active and reactive powers of branches, voltage amplitudes of buses, electricity powers generated by coal-fired unit and wind turbine, and also the powers purchased from the bulk power grid in power system; (2) the temperatures of hot water and insulation layer, the heat exchange power at the heat exchange station, and heat loads in heating network; (3) the gas, heat and electric powers of the CHP unit.

The optimization problem of end-users formulated in Section 4.2 is summarized as follows:

$$\begin{cases} \min C_n^{\text{U}} \\ \text{s.t.} (35) \text{ - } (36), (38) \text{ - } (39), (41) \text{ - } (49) \end{cases} \tag{51}$$

The optimization variables encompass the shiftable and convertible loads, indoor temperatures and the PMV indices of end-users.

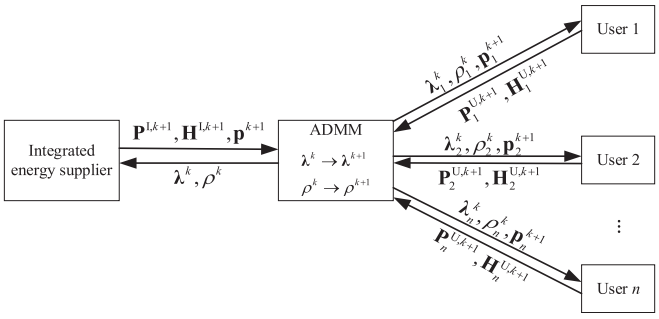


Fig. 6. Framework of the distributed interactive scheduling approach using ADMM-VPP.

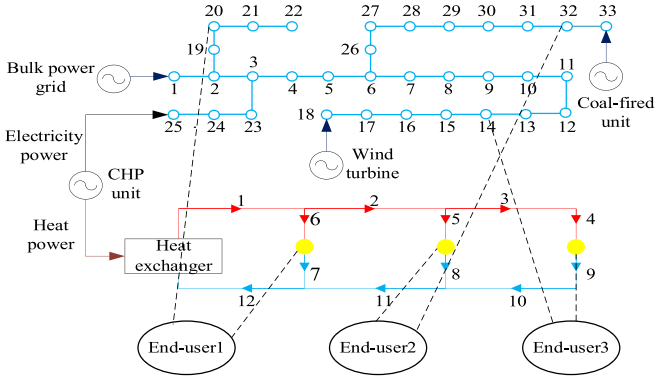


Fig. 7. Topology of IES.

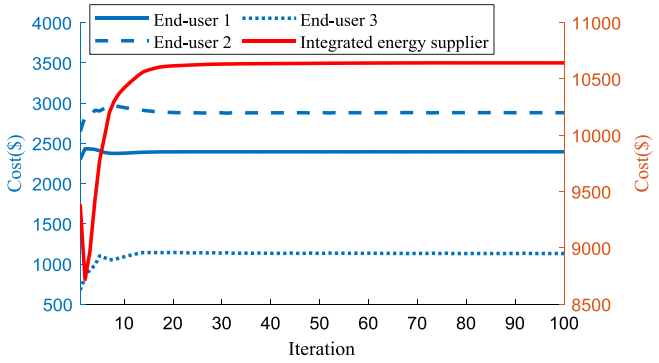


Fig. 8. Optimization objectives of participants during iterations.

5. ADMM-VPP based distributed interactive scheduling approach

The integrated energy supplier and users, as independent individuals, prioritize their own interests. Moreover, the privacy of individuals should be protected from each other. Only the necessary information can be exchanged among these entities to maintain individual privacy. To address these challenges, ADMM-VPP is employed to achieve the coordinated scheduling optimization between the integrated energy supplier and users.

The coupling variables between the supplier and users are the interactive powers of electricity and heat energy, which should be decoupled respectively for the supplier and users, denoted by the superscript "I" and "U". After the kth round of information interaction and decision adjustment, the consistency between the sold powers by the supplier and the purchased powers by users can be reached, described as:

$$\begin{aligned} P_{n,t}^{\mathrm{I},k} &= P_{n,t}^{\mathrm{U},k} \\ H_{n,t}^{\mathrm{I},k} &= H_{n,t}^{\mathrm{U},k} \end{aligned} \tag{52}$$

The ADMM method transforms the original objective function into the augmented Lagrangian. The optimization models of the supplier and user n are reformulated as:

$$\begin{aligned} \min \mathbb{L}^{\mathrm{I},k+1} = C^{\mathrm{I}} &+ \sum_{t\in\mathscr{T}}\sum_{n\in\mathscr{N}} \lambda_{\mathrm{e},n,t}^{k}(P_{n,t}^{\mathrm{I}} - P_{n,t}^{\mathrm{U},k}) + \sum_{t\in\mathscr{T}}\sum_{n\in\mathscr{N}} \lambda_{\mathrm{h},n,t}^{k}(H_{n,t}^{\mathrm{I}} - H_{n,t}^{\mathrm{U},k}) + \\ &\sum_{t\in\mathscr{T}}\sum_{n\in\mathscr{N}} \frac{\rho_{\mathrm{e},n,t}}{2}\left\|P_{n,t}^{\mathrm{I}} - P_{n,t}^{\mathrm{U},k}\right\|_2^2 + \sum_{t\in\mathscr{T}}\sum_{n\in\mathscr{N}} \frac{\rho_{\mathrm{h},n,t}}{2}\left\|H_{n,t}^{\mathrm{I}} - H_{n,t}^{\mathrm{U},k}\right\|_2^2 \end{aligned} \tag{53}$$

$$\begin{aligned} \min \mathbb{L}_{n}^{\mathrm{U},k+1} = C_{n}^{\mathrm{U}} &+ \sum_{t\in\mathscr{T}} \lambda_{\mathrm{e},n,t}^{k}(P_{n,t}^{\mathrm{I},k+1} - P_{n,t}^{\mathrm{U}}) + \sum_{t\in\mathscr{T}} \lambda_{\mathrm{h},n,t}^{k}(H_{n,t}^{\mathrm{I},k+1} - H_{n,t}^{\mathrm{U}}) + \\ &\sum_{t\in\mathscr{T}} \frac{\rho_{\mathrm{e},n,t}}{2}\left\|P_{n,t}^{\mathrm{I},k+1} - P_{n,t}^{\mathrm{U}}\right\|_2^2 + \sum_{t\in\mathscr{T}} \frac{\rho_{\mathrm{h},n,t}}{2}\left\|H_{n,t}^{\mathrm{I},k+1} - H_{n,t}^{\mathrm{U}}\right\|_2^2 \end{aligned} \tag{54}$$

where $\lambda_{\mathrm{e},n,t}^{k}$ and $\lambda_{\mathrm{h},n,t}^{k}$ are the Lagrangian multipliers of electricity and heat powers of user n at the iterative step k, $\rho_{\mathrm{e},n,t}$ and $\rho_{\mathrm{h},n,t}$ are the penalty parameters.

The Lagrangian multipliers are updated according to:

$$\begin{aligned} \lambda_{\mathrm{e},n,t}^{k+1} &= \lambda_{\mathrm{e},n,t}^{k} + \rho_{\mathrm{e},n,t}(P_{n,t}^{\mathrm{I},k+1} - P_{n,t}^{\mathrm{U},k+1}) \\ \lambda_{\mathrm{h},n,t}^{k+1} &= \lambda_{\mathrm{h},n,t}^{k} + \rho_{\mathrm{h},n,t}(H_{n,t}^{\mathrm{I},k+1} - H_{n,t}^{\mathrm{U},k+1}) \end{aligned} \tag{55}$$

The convergence of ADMM algorithm is enforced by both the primal residual and dual residual, expressed as:

$$\begin{aligned} \left\|\mathscr{R}^{k}\right\|_2^2 &= \sum_{t\in\mathscr{T}}\sum_{n\in\mathscr{N}}\left(\left\|P_{n,t}^{\mathrm{I},k} - P_{n,t}^{\mathrm{U},k}\right\|_2^2 + \left\|H_{n,t}^{\mathrm{I},k} - H_{n,t}^{\mathrm{U},k}\right\|_2^2\right) \\ \left\|\mathscr{S}^{k}\right\|_2^2 &= \sum_{t\in\mathscr{T}}\sum_{n\in\mathscr{N}}\left(\left\|\left(-\rho_{\mathrm{e},n,t}\right)\left(P_{n,t}^{\mathrm{I},k} - P_{n,t}^{\mathrm{I},k-1}\right)\right\|_2^2 + \left\|\left(-\rho_{\mathrm{h},n,t}\right)\left(H_{n,t}^{\mathrm{I},k} - H_{n,t}^{\mathrm{I},k-1}\right)\right\|_2^2\right) \end{aligned} \tag{56}$$

Once both the primal and dual residuals fall below the pre-defined stopping threshold ε, all entities stop their individual iterations, ensuring global optimization coordination.

In traditional ADMM, ρ is generally fixed. However, different values have a great influence on the iteration. The larger ρ imposes a stricter penalty on primal feasibility violations, yielding smaller primal residuals. Conversely, the definition of dual residual suggests that the smaller ρ tends to reduce the dual residual, resulting in a larger primal residual. Consequently, ADMM-VPP is proposed. The penalty parameter ρ adapts in each iteration based on the relative sizes between the primal and dual residuals, shown as follows:

$$\boldsymbol{\rho}^{k+1} := \begin{cases} \tau^{\mathrm{incr}}\boldsymbol{\rho}^{k} & \text{if } \left\|\mathscr{R}^{k}\right\|_2^2 > \mu\left\|\mathscr{S}^{k}\right\|_2^2 \\ \boldsymbol{\rho}^{k}/\tau^{\mathrm{decr}} & \text{if } \left\|\mathscr{S}^{k}\right\|_2^2 > \mu\left\|\mathscr{R}^{k}\right\|_2^2 \\ \boldsymbol{\rho}^{k} & \text{otherwise} \end{cases} \tag{57}$$

where the parameters $\mu > 1$, $\tau^{\mathrm{incr}} > 1$, $\tau^{\mathrm{decr}} > 1$. A common practice is to set $\mu = 10$ and $\tau^{\mathrm{incr}} = \tau^{\mathrm{decr}} = 2$ [41]. Through updating $\boldsymbol{\rho}$, the penalty parameter tries to keep primal and dual residual norms within the factor μ of one another as they both converge to zero. The convergence performance of ADMM is improved and the dependence of the algorithm on the initial value of $\boldsymbol{\rho}$ is weakened. The framework of distributed interactive scheduling approach using ADMM-VPP is displayed in Fig. 6.

The procedure of the distributed interactive schedule using ADMM-VPP is illustrated as follows:

1. **Initialization:**Set the initial values of $\lambda_{\mathrm{e},n,t}^{1}$,$\lambda_{\mathrm{h},n,t}^{1}$ and $\rho_{\mathrm{e},n,t}^{1}$, $\rho_{\mathrm{h},n,t}^{1}$. Users report their energy demands to the supplier.
2. **While** $k \leqslant k_{\max}$ and $\max(\left\|\mathscr{R}^{k}\right\|_2^2, \left\|\mathscr{S}^{k}\right\|_2^2) \geqslant \varepsilon$, **do:**

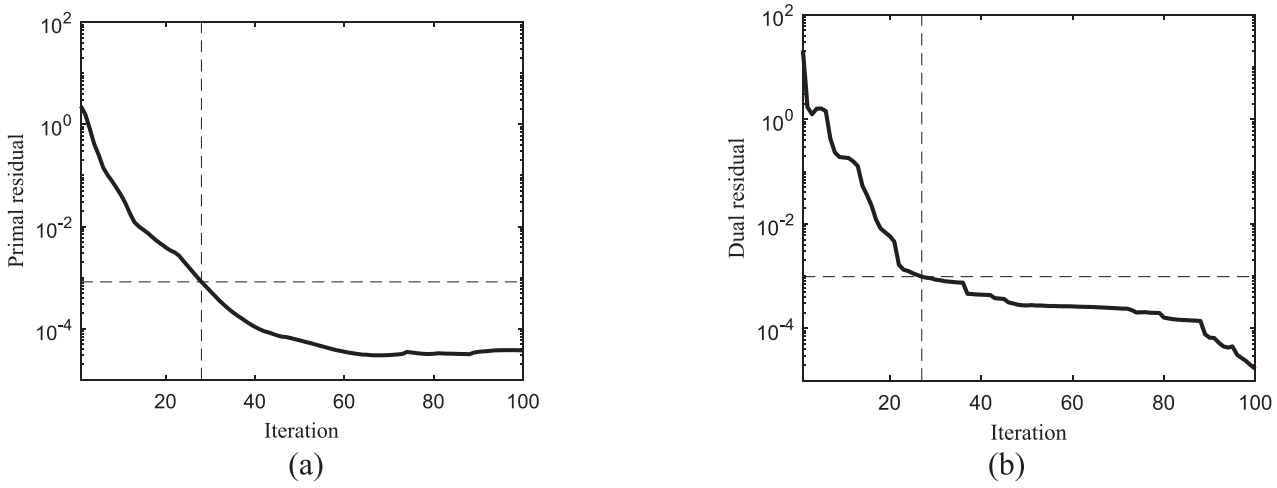


Fig. 9. (a) Primal residuals and (b) dual residuals during iterations.

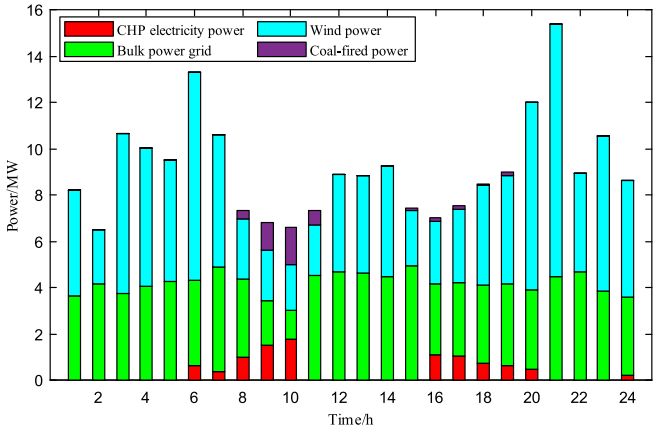


Fig. 10. Scheduling results of energy generations of the supplier.

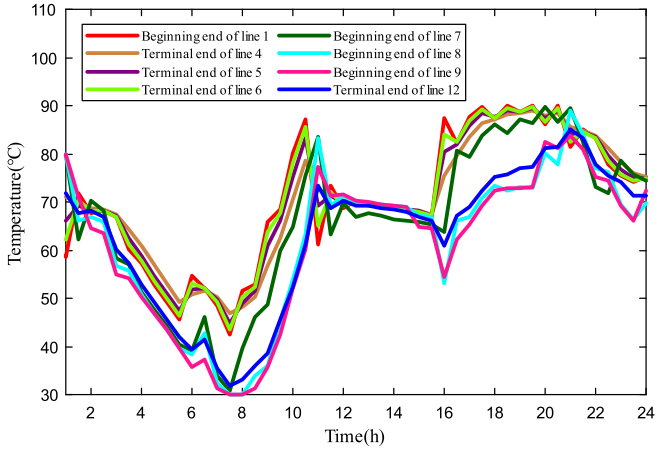


Fig. 11. Water temperatures in heating network.

3. The supplier conducts low-carbon economic scheduling based on the augmented Lagrangian, with the optimization objective in Eq. (14) and the constraints in Eqs. (15)-(34).
4. The supplier determines users' energy-carbon prices according to Section 3.2 and transmits the pricing results to the corresponding user.
5. **for** each user n:

Table 1
Carbon prices of end-users.

	Electric Load ($/MWh)				Heat Load ($/MWh)
	Basic load	Flexible load			
		Valley	Plain	Peak	
End-user 1	36.973	31.113	31.406	20.950	19.030
End-user 2	40.573	30.394	32.125	31.150	19.754
End-user 3	15.006	11.109	8.157	12.388	20.595

Table 2
Energy prices of end-users.

	Electric load ($/MWh)			Heat load ($/MWh)
	Valley	Plain	Peak	
End-user 1	48.042	80.763	163.609	156.787
End-user 2	45.337	75.050	140.573	157.591
End-user 3	16.767	41.442	60.931	159.316

6. User n carries out IDR to respond to the energy-carbon price, with the optimization objective in Eq. (37) and the constraints in Eqs. (35)-(36),(38)-(39), (41)-(49). The updated electricity and heat load demands are transmitted back to the supplier.
7. **end for**
8. Update the Lagrangian multipliers $\lambda_{e,n,t}^{k+1}$ and $\lambda_{h,n,t}^{k+1}$ according to Eq. (55). And update the penalty parameters $\rho_{e,n,t}^{k+1}$ and $\rho_{h,n,t}^{k+1}$ based on Eq. (57). Return to step 2 until the convergence is achieved.
9. **end while**

6. Case study

To demonstrate the efficacy of the proposed algorithm, an IES comprising one supplier and three end-users is established in Fig. 7. The integrated energy supplier delivers the energy through an IEEE 33-bus standard distribution network and a 12-pipeline heating network to three end-users. The electricity and heat networks are interconnected via a CHP unit. The IES interfaces with the bulk power grid at node 1, a wind turbine at node 18 and a coal-fired unit at node 33. Parameters for the simulations are given in Tables A1-A4 and Figs. A1-A5 [12,26,34]. Basic, shiftable and convertible loads contribute to 70 %, 20 % and 10 % of each user's load respectively. The scheduling horizon is chosen as 24 h, with the time steps $\Delta t = 1$ h and $\Delta\tau = 30$ min. The stopping thresholds ε in ADMM-VPP is set to 10^{-3}. All the following simulations are executed

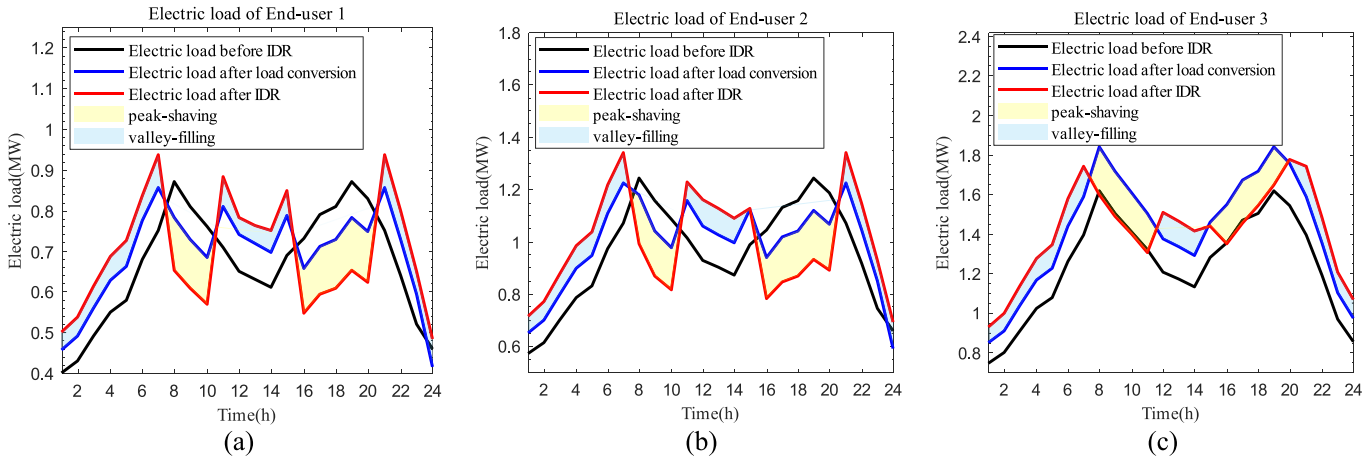


Fig. 12. Electric load curves of (a) End-user 1, (b) End-user 2, and (c) End-user 3 before and after IDR.

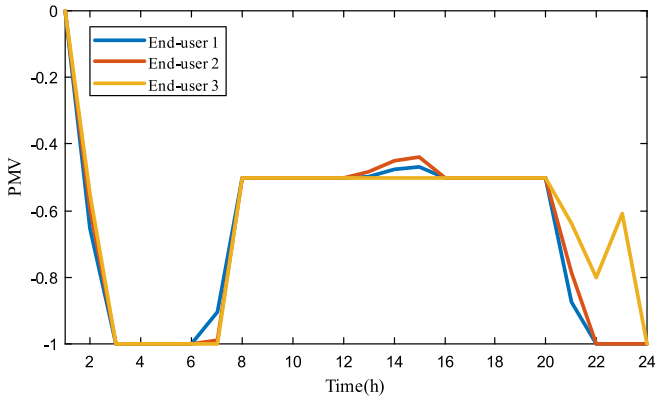


Fig. 13. PMV of users after IDR.

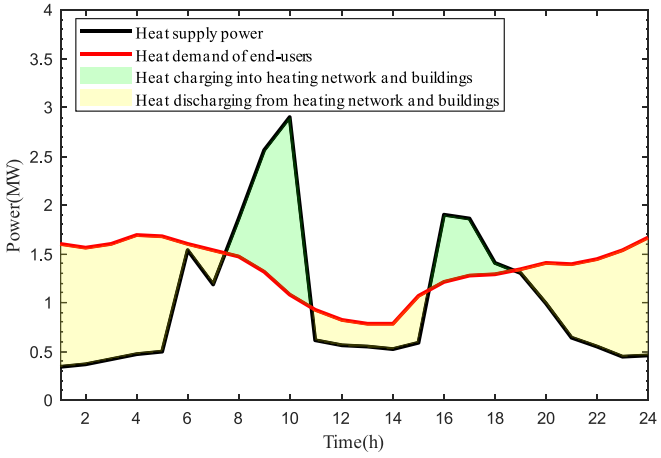


Fig. 14. Heat energy charging and discharging.

using the Gurobi solver in MATLAB on a computer equipped with Intel Core i7-9700 CPU and 32 GB RAM.

6.1. *Analysis of coordinated scheduling optimization results*

The ADMM-VPP approach facilitates the coordination of scheduling between the integrated energy supplier and end-users. During the iteration process, the optimization objectives of all participants are shown in Fig. 8. The primal residual and dual residual are drawn in Fig. 9. It can be seen that at the beginning of the iterations, each participant swiftly modifies its scheduling strategy after interactions with others. The optimization objective of the supplier initially decreases and subsequently rises, whereas for end-users, the pattern is reversed. After approximately 28 iterations, both the primal and dual residuals achieve the stipulated convergence criteria.

Table 3
Comparison on energy-carbon costs of Cases 1-A, 1-B and 1-C.

	C^U ($)			C^I ($)	Total ($)	Carbon Emissions (tCO_2)
	End-user 1	End-user 2	End-user 3	Supplier		
Case 1-A	2,395	2,878	1,131	10,641	17,045	69.60
Case 1-B	3,320	3,784	2,507	13,689	23,300	69.60
Case 1-C	3,606	4,041	2,762	14,199	24,608	69.60

The energy generation scheduling results of the supplier is depicted in Fig. 10. During the scheduling period, wind power is fully consumed. During peak hours, the more cost-effective coal-fired and CHP units are prioritized for higher electric power generation. Additionally, owing to the heat-led operational mode of the CHP unit, its heat production also aligns with peak-load hours. As illustrated in Figs. 10 and 11, the CHP unit generates power between 6-10 h and 16–20 h. And the water temperatures of heating network elevate between 7-11 h and 16–21 h, attributable to the delay in heat transfer. To cater to the heating demands throughout the scheduling period, the intrinsic storage properties of the heating network are leveraged.

The trading energy-carbon prices are listed in Tables 1 and 2. The energy-carbon price of the heat load is correlate with the distance between the heat load and the heat exchange station, for longer transmission distances can result in increased heat loss. There's a pronounced correlation between the energy-carbon price of the electric load and its location within the electric network. Being in closer proximity to the wind turbine, End-user 3 has the lowest carbon and electricity prices. While the carbon price of End-user 2′s flexible load at peak time surpasses that of End-user 1, the electricity price exhibits the inverse trend. The rationale is illustrated as follows: End-user 2 is situated near the coal-fired generator, whereas End-user 1 is proximal to both the bulk power grid and the CHP unit. During peak hours, the cost-effective but albeit polluting coal-fired unit results in a higher carbon price but lower electricity prices for End-user 2.

The electric load curves both before and after the implementation of IDR are drawn in Fig. 12. The black curve represents the original electric load prior to IDR. The blue curve represents the electric load after

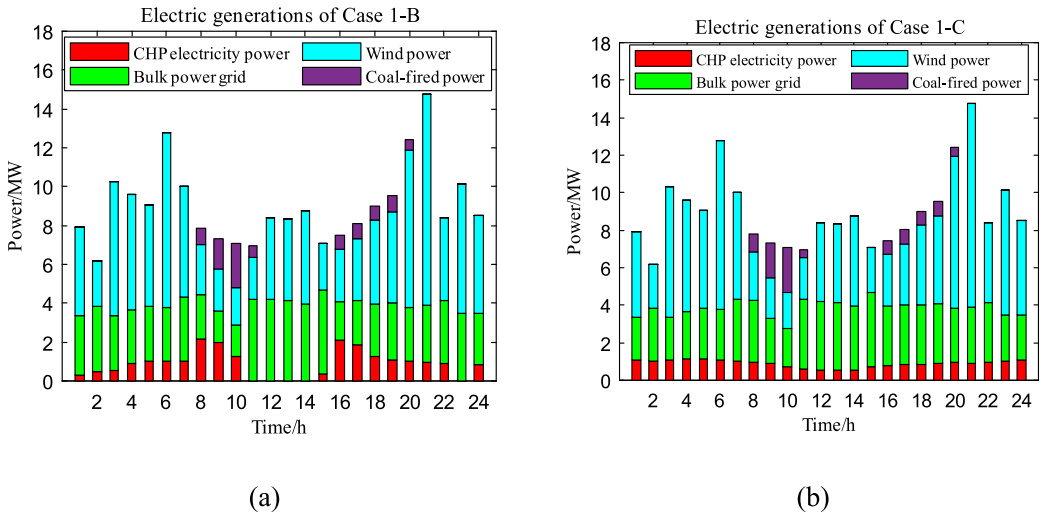


Fig. 15. Scheduling results of energy generations in (a) Case 1-B and (b) Case 1-C.

Table 4
Comparison on energy-carbon costs of Cases 2-A, 2-B, 2-C and 2-D.

	C^I ($)			Carbon emissions (tCO_2)
	Generation cost	Carbon cost	Total cost	
Case 2-A	10,061	580	10,641	69.60
Case 2-B	9,487	0	9,487	101.45
Case 2-C	9,899	756	10,655	73.11
Case 2-D	10,112	1,093	11,205	68.93

electricity-heat conversion. And the red curve reflects the state after IDR, showcasing further electric load shifts subsequent to the conversion. As shown in Table 2, during peak periods, the electricity energy consumption of End-users 1 and 2 is less cost-efficient than heat energy utilization, after considering the electricity-heat conversion ratio. Therefore, heat energy demand is satisfied through heat network. During non-peak periods, however, the heating demand is met by using cheaper electric power. For End-user 3, because its electricity price significantly undercuts heat price, there's an uptick in electric power usage to meet the heating demand. Comparing the red and blue curves reveals that, peak-time electric loads can undergo further reduction through load shifting. The simulation results demonstrate that the implementation of IDR can effectively harness the end-users' flexibility.

The PMV values after carrying out IDR are depicted in Fig. 13. The PMV value falls within the comfort range, ensuring no discomfort for end-users. The heat storage effect of heating network and buildings is

Table 5
Comparison on energy-carbon costs of Cases 3-A, 3-B and 3-C.

	C^U ($)			C^I ($)	Carbon emissions (tCO_2)
	User 1	User 2	User 3	Supplier	
Case 3-A	2,395	2,878	1,131	10,641	69.60
Case 3-B	1,754	2,024	2,434	10,843	72.96
Case 3-C	927	2,985	4,168	11,856	73.81

Table 6
Energy-carbon prices in Case 3-B.

Energy price ($/MWh)				Carbon Price ($/MWh)
Electric Load			Heat Load	
Valley	Plain	Peak		
28.580	49.217	104.419	180.529	4.807

Table 7
Carbon prices of end-users in Case 3-C.

	Electric Load ($/MWh)				Heat Load ($/MWh)
	Basic load	Flexible load			
		Valley	Plain	Peak	
End-user 1	11.363	7.199	13.376	11.258	17.880
End-user 2	39.187	31.759	37.478	35.791	19.630
End-user 3	38.699	38.000	29.242	27.773	19.650

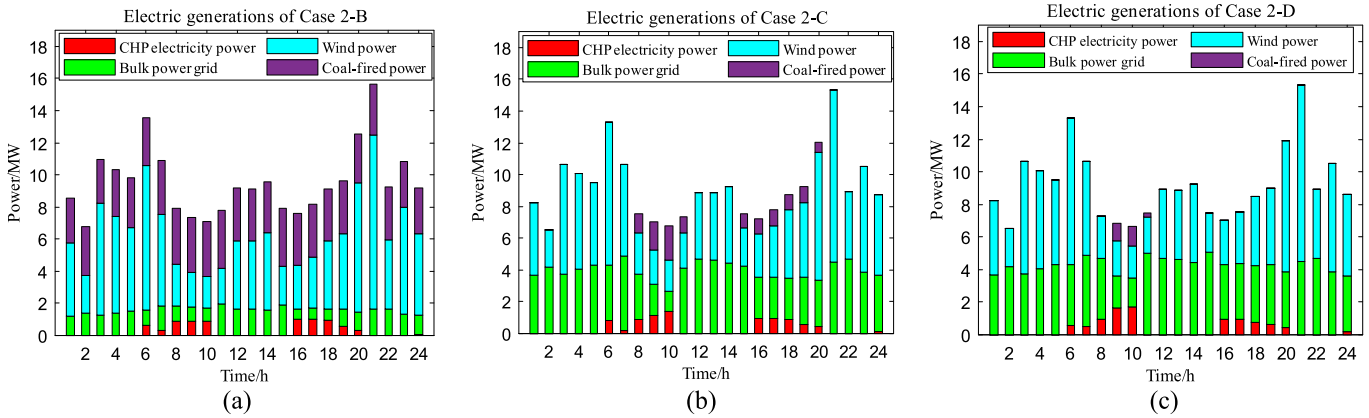


Fig. 16. Scheduling results of energy generation in (a) Case 2-B, (b) Case 2-C and (c) Case 2-D.

Table 8
Energy prices of end-users in Case 3-C.

	Electric Load ($/MWh)			Heat Load ($/MWh)
	Valley	Plain	Peak	
End-user 1	29.122	31.798	63.172	159.396
End-user 2	54.470	76.296	143.681	157.745
End-user 3	47.423	104.164	166.630	154.447

shown in Fig. 14. During peak periods, CHP unit increases its output, and the surplus heat energy is stored in both buildings and heating network. Conversely, during non-peak periods, the pre-stored heat energy is released to satisfy heat demand. The above results indicate that the thermal storage effect and IDR can enhance system flexibility, subsequently lowering operating costs for both the supplier and end-users.

6.2. Analysis of system flexibility effect

In order to investigate the effect of system flexibility on the energy-carbon costs of the supplier and end-users, this section presents a comparison across the following three cases:

Case 1-A: The scenario in Section 6.1.

Case 1-B: Solely the thermal storage characteristics of the heating network are utilized. The IDR of end-users is not considered. And the heat load is chosen as the heat demand when PMV is zero.

Case 1-C: Neither the thermal storage capacity of heating network nor the IDR of end-users is considered. The heat load is the same as Case 1-B. The heat output of CHP unit and heat load are balanced in real time.

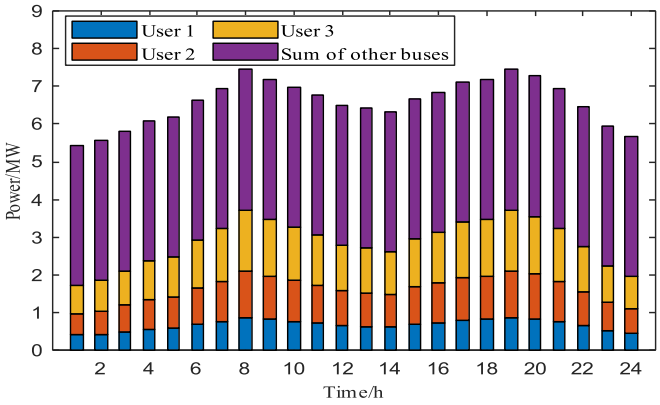


Fig. A1. Electric load in power system.

The comparison on the energy-carbon costs of the three cases is shown in Table 3. Considering the storage effect of heating network and IDR, costs for both the supplier and end-users are reduced. The total cost of Case 1-A is about 30.73 % lower than that of Case 1-C. The generation scheduling results of Case 1-B and 1-C are displayed in Fig. 15. Compared with the scheduling results of Case 1-B and 1-C, the CHP generation of Case1-A shown in Fig. 10 is smaller. This suggests that, owing to IDR, a greater portion of the heat load is converted to electric

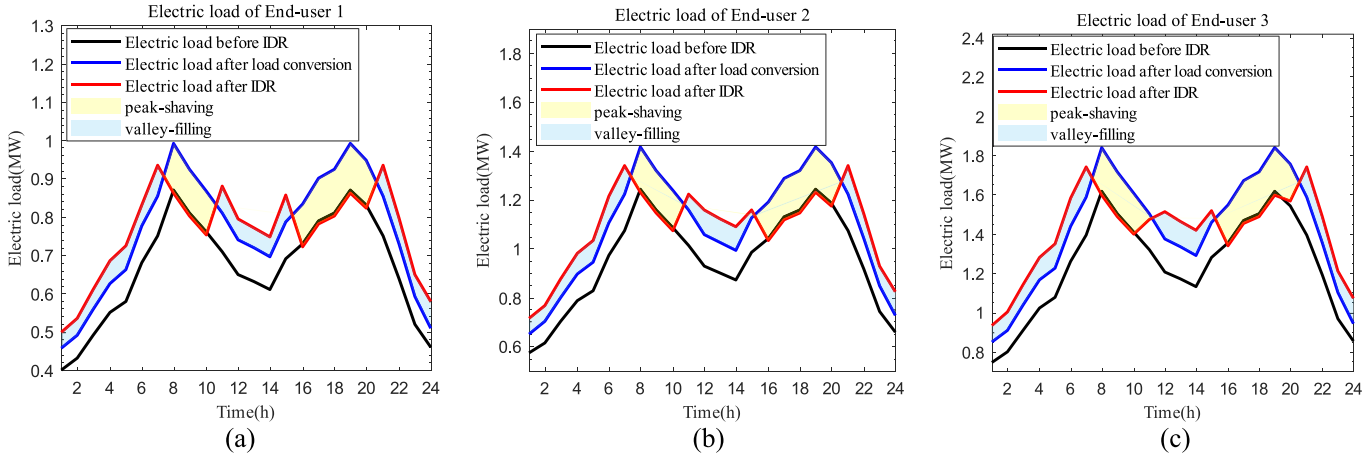


Fig. 17. Electric load curves of (a) End-user 1 (b) End-user 2 (c) End-user 3 before and after IDR in Case 3-B.

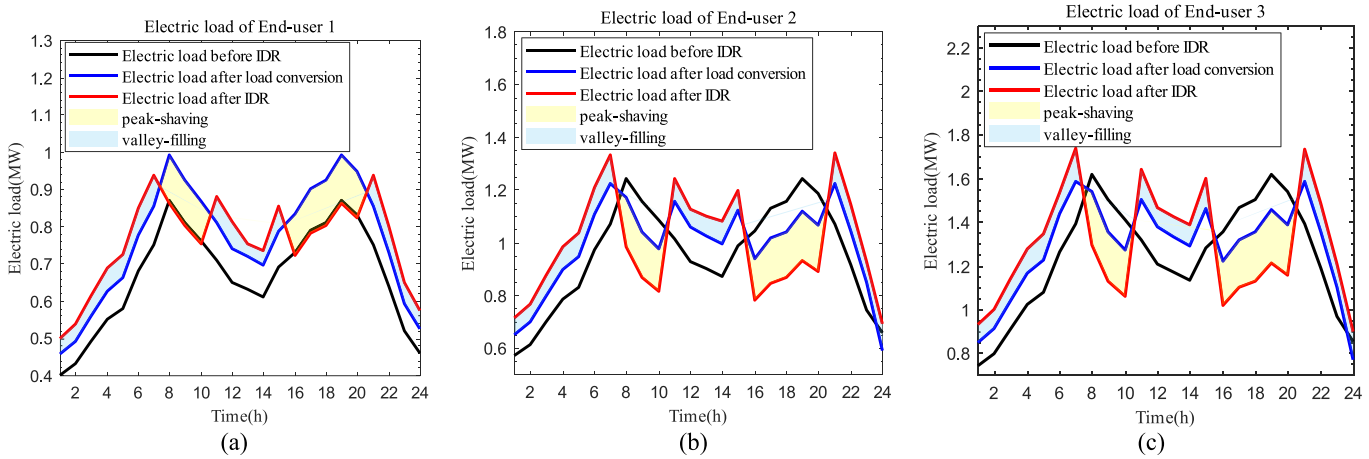


Fig. 18. Electric load curves of (a) End-user 1, (b) End-user 2 (c), and End-user 3 after IDR in Case 3-C.

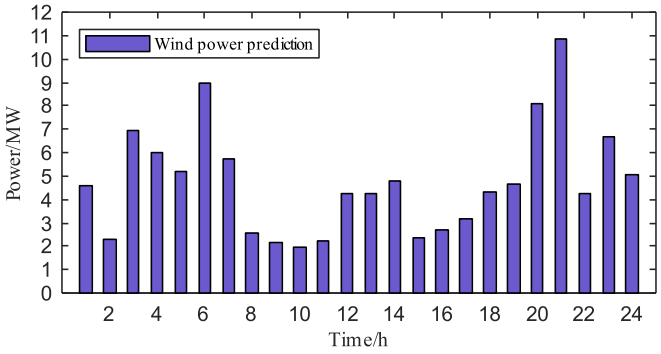


Fig. A2. Wind power prediction value.

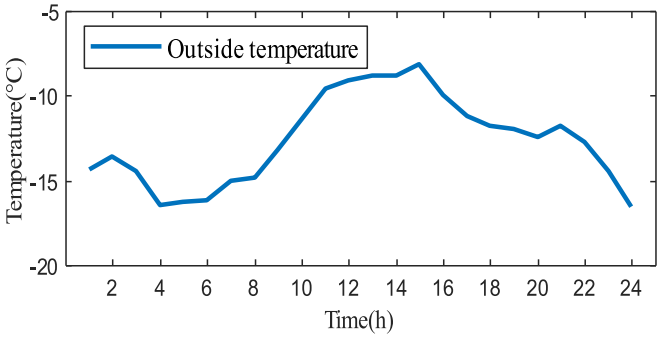


Fig. A3. Outside temperature.

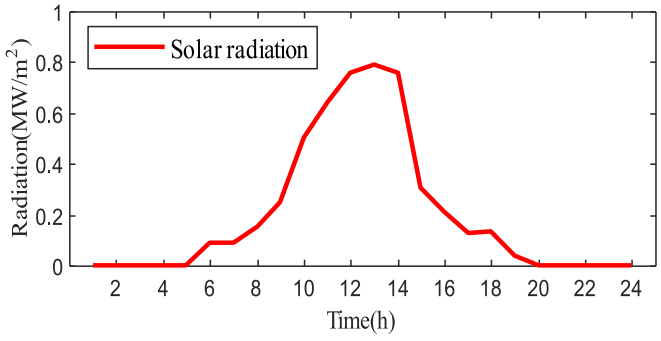


Fig. A4. Solar radiation.

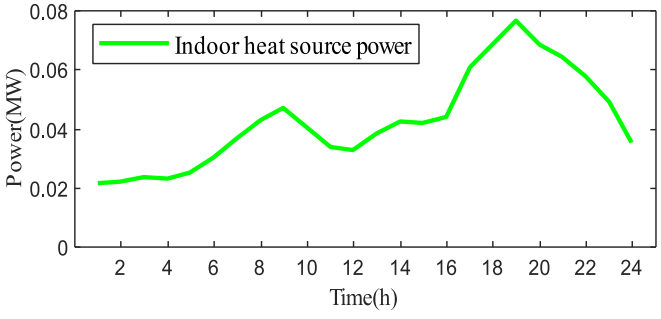


Fig. A5. Indoor heat source power.

load, with heat demand met by directly consuming electrical energy at the end-user's side. Compared to Cases 1-A and 1-B, the CHP generation in Case 1-C is more even. While the CHP unit operates in the heat-led mode, the system's economy benefits from the thermal storage effect of the heating network. For example, the energy-carbon cost of Case 1-B is reduced by 5.32 %, compared with Case 1-C.

Table 9
Comparison on the costs of the centralized and distributed optimizations.

	C^U ($)			C^I ($)	Total ($)	Total Error
	User 1	User 2	User 3	Supplier		
Centralized optimization	2,395	2,878	1,131	10,641	17,045	0.28 %
Distributed optimization	2,391	2,875	1,115	10,617	16,998	0 %

Table 10
Comparison of standard ADMM and ADMM-VPP.

	ρ_P/ρ_H	Iteration count	Iteration duration (s)
Standard ADMM	40/20	52	5104
	80/40	48	4573
	160/80	63	6119
	320/160	84	8140
ADMM-VPP	40/20	33	2832
	80/40	28	2524
	160/80	40	3628
	320/160	63	5700

Table A1
Simulation parameters.

Parameters	Value	Parameters	Value
Δx(m)	50	ΔT_{hn}^{max}(°C)	5
l_{max}(pu)	1100	ΔT_{hn}^{min}(°C)	−5
v_{max}(pu)	1.14	T_{hn}^{max}(°C)	90
v_{min}(pu)	0.8649	T_{hn}^{min}(°C)	30
P_g^{down}(MW·(min)$^{-1}$)	−1.33	ρ_u/(kg·m^{-3})	1000
P_g^{up}(MW·(min)$^{-1}$)	1.33	K_e	0.3
D_{in}(m)	0.4	K_h	0.45
D_{out}(m)	0.6	P_{CHP}^{down}(MW·(min)$^{-1}$)	−1.33
h(W·(m^2·K)$^{-1}$)	7000	P_{CHP}^{up}(MW·(min)$^{-1}$)	1.33
ξ_u(W·(m·K)$^{-1}$)	0.31	η_{P2H}	0.83
ξ_s(W·(m·K)$^{-1}$)	0.033	η_{ex}	1
c_r(J·(kg·°C)$^{-1}$)	1380	η_{load}	1
ρ_r/(kg·m^{-3})	1000	d_{trans}($)	0.2
c_u(J·(kg·°C)$^{-1}$)	4200	d_{conv}($)	0.05
M_4 (kg·s^{-1})	6	A_n^{fc}(MW·(m^2·K)$^{-1}$)	0.15
M_5 (kg·s^{-1})	6	S_n^{fc}(m^2)	1
M_6 (kg·s^{-1})	6	α_n	0.4
Z (m)	1	S_n^{win}(m^2)	0.8
F_n (MW·(m^2)$^{-1}$)	500	e_{profit}	1.2
I (W/m^2)	70	T_{comfy}(°C)	33.5
R_{clo} ((m^2·°C)/W)	0.11		

Table A2
Time-of-use price of bulk power grid and coal-fired unit.

Time periods	Price of bulk power grid/coal-fired unit ($/MWh)
Valley: 1–5, 22–24	38/30
Plain: 6–7, 11–15,21	62/50
Peak: 8–10, 16–20	132/106

It is noteworthy that the carbon emissions remain unchanged after leveraging the system's flexibility. Simulation observations reveal that the carbon emissions trading volume of the scheduling results aligns with the upper limit of the first-tier carbon price. If the carbon emissions increase further, the carbon cost will show a stepwise growth. Consequently, to optimize operational economy, it becomes imperative to reduce generation from higher pollution units, like the coal-fired generator, ensuring carbon emissions remain within the first tier.

6.3. Analysis of LCTM

Four cases are further researched to explore the influence of LCTM on

Table A3
Parameters of generation units.

Type	Wind turbine	Coal-fired unit	Bulk power grid	CHP unit
Minimum output (MW)	0	0	0	0
Maximum output (MW)	Prediction output	30	/	45(heat)/30 (electricity)
Price coefficient ($/MWh)	0	in Table A2	in Table A2	100
Carbon coefficient (tCO_2/MWh)	0	1	0.647	0.2

Table A4
Parameters of LCTM.

Carbon quota	Interval division	Ladder price
$E_{quota} = 58$	$\sigma_1 = 0.2, \sigma_2 = 0.5$	$\lambda_1 = 50, \lambda_2 = 80, \lambda_3 = 120$

scheduling:

Case 2-A: The scenario in Section 6.1.

Case 2-B: Carbon emission cost is excluded during scheduling.

Case 2-C: Carbon price λ_1 remains consistent throughout the scheduling period, with no ladder structure.

Case 2-D: Implements a ladder-type carbon price, twice that of Case 2-A.

The costs of energy generation and carbon emissions for Cases 2-B, 2-C and 2-D are listed in Table 4. The scheduling results of electric generation are depicted in Fig. 16. In Case 2-B, there's a notable reliance on the inexpensive but polluting coal-fired generator, leading to significant pollution. Owing to LCTM in Case 2-A, the carbon emissions decrease by 31.39 % compared to Case 2-B. When compared with Case 2-C, carbon emissions in Case 2-A exhibit a 4.8 % reduction, all while incurring lower carbon emission and total energy-carbon costs, highlighting the efficacy of LCTM. In Case 2-D, the amplified ladder price of LCTM results in decreased coal-fired generation and marginally reduced carbon emissions, but a notable uptick in carbon trading costs. Therefore, establishing a reasonable ladder-type carbon pricing mechanism is pivotal to efficiently curbing carbon emissions without unduly burdening users financially.

6.4. Analysis of energy-carbon cost-sharing mechanism

In order to assess the influence of cost-sharing mechanism on carbon emissions and operation costs, three cases are compared:

Case 3-A: The scenario in Section 6.1.

Case 3-B: Each end-user is assigned a unified energy-carbon price derived by dividing the total energy-carbon cost by the collective load. Additionally, the allocated energy price is multiplied by the profit rate to determine the final energy purchase price.

Case 3-C: The topological positions of End-user 1 and End-user 3 are interchanged.

The energy-carbon costs associated with the three cases are given in Table 5. The corresponding energy-carbon prices are listed in Tables 6-8. The electric loads after IDR are drawn in Fig. 17 and Fig. 18. In Case 3-B, owing to a uniform pricing strategy, each user tends to adopt a congruent IDR approach. For this case, the energy-carbon costs of End-users 1 and 2, respectively proximate to the bulk power grid and coal-fired generation, decrease by 26.76 % and 29.67 %. Conversely, the energy-carbon cost of End-user 3, positioned near the eco-friendly and economical wind power, escalates by 115.21 %. Compared with Case 3-A, this pricing strategy fails to accurately capture each user's contribution to the system's carbon emissions and operating costs. It consequently diminishes the incentive for end-users to establish on-site renewable energy sources. Furthermore, when compared with Case 3-A, the total energy-carbon cost and carbon emissions in Case 3-B increase by 1.90 % and 4.83 % respectively. This is attributed to the fact that the differential pricing aids end-users in executing precise IDR to mitigate both energy costs and carbon emissions.

In Case 3-C, the energy-carbon prices of End-user 2 remain relatively stable. However, those of End-user 1 decrease, while those of End-user 3 increase, compared to the energy-carbon prices of Case 3-A listed in Tables 1 and 2. This discrepancy arises because End-user 1 is now in closer proximity to wind turbine, instead of End-user 3. Fig. 17 further reveals that, the electricity load patterns of End-user 1 after IDR in Case 3-C closely resemble those of End-user 3 in Case 3-A. That is, a more competitive electricity price prompts a greater heat demand to be met through direct electricity consumption. And the patterns of End-user 3 after IDR align with those of End-user 1 in Case 3-A. The above results illustrate the equity and efficacy of the hierarchical Shapley-based allocation approach proposed.

6.5. Analysis of distributed interactive scheduling method

To validate the efficiency of ADMM-VPP, the cases of the distributed and centralized optimization are compared. The scenario outlined in Section 6.1 represents the distributed case. In centralized optimization, the sum of optimization objectives of supplier and end-users is chosen as the optimization objective. The energy-carbon prices in Tables 1 and 2 are used for this optimization. A comparison on the costs of the two cases is listed in Table 9.

Under centralized scheduling, the supplier and end-users can collect detailed information of each other. Conversely, in the framework of distributed coordinated scheduling, only essential details related to energy purchasing and pricing are exchanged, ensuring the privacy of participants. A discrepancy of approximately 0.28 % between centralized and distributed scheduling outcomes signifies the reliability of the ADMM-based distributed coordinated scheduling approach.

In order to show the superiority of ADMM-VPP in contrast to the standard ADMM, the same stop threshold and initial penalty parameters are maintained. A comparison of iteration duration and count is shown in This underscores the proven efficacy of ADMM-VPP.

Table 10. Compared with the standard ADMM, the average convergence time is curtailed by 38.65 %, and the average iteration count is decreased by 33.60 %. This underscores the proven efficacy of ADMM-VPP.

7. Conclusion

In order to reduce carbon emissions, thereby mitigating climate change, a distributed interactive optimization approach consisting of an integrated energy supplier and multi-end-users is proposed, in which a hierarchical Shapley based energy-carbon pricing strategy is developed to reasonably and fairly share the energy-carbon costs among end-users. By employing ADMM-VPP, the coordinated scheduling optimization among the supplier and end-users can be achieved, with only the energy demand and price being exchanged. The energy-carbon pricing strategy serves as the linkage between the CET market and end-users, encouraging end-users to conduct IDR and build on-site renewable energy sources, so as to facilitate participation of end-users in the CET market. In the proposed approach, the thermal storage effect of the supplier's heating network, users' buildings and IDR are fully exploited to reduce the overall cost. The reasonable ladder-type pricing in CET market is more helpful for reducing carbon emissions, without imposing additional financial burden on energy consumers. ADMM-VPP facilitates the scheduling optimization for multi-participants while safeguarding their privacy, showcasing superior performance to the standard ADMM algorithm both in terms of the stronger convergence and reduced sensitivity to the initial value of the penalty parameter. The methodology proposed in this study is congruent with the prevailing landscape of CET markets, which principally engages large-scale industrial users through

a reward-penalty-based LCTM. Consequently, it harbors substantial potential for application in forthcoming scenarios. Our future work will investigate the interaction between the supplier and end-users where an interest alliance among end-users exists.

CRediT authorship contribution statement

Ruijie Liu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing. **Zhejing Bao:** Conceptualization, Methodology, Validation, Formal analysis, Supervision, Project administration, Funding acquisition. **Zhenting Yu:** Methodology. **Chenjian Zhang:** Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work is supported by the National Key R&D Program of China (2022YFB3304502).

Appendix A

(See Table A1-A4).

References

[1] Ahonen E, Corbet S, Goodell JW, Günay S, Larkin C. Are carbon futures prices stable? New evidence during negative oil. Financ Res Lett Jun. 2022;47:102723. https://doi.org/10.1016/j.frl.2022.102723.
[2] Convery FJ, Redmond L. Market and price developments in the European union emissions trading scheme. Rev Environ Econ Policy 2007;1(1):88–111. https://doi.org/10.1093/reep/rem010.
[3] Richter JL, Mundaca L. Market behavior under the New Zealand ETS. Carbon Manage Aug. 2013;4(4):423–38. https://doi.org/10.4155/cmt.13.33.
[4] Shabanpour-Haghighi A, Seifi AR. Multi-objective operation management of a multi-carrier energy system. Energy Aug. 2015;88:430–42. https://doi.org/10.1016/j.energy.2015.05.063.
[5] Xiang Y, et al. Low-carbon economic dispatch of electricity-gas systems. Energy Jul. 2021;226:120267. https://doi.org/10.1016/j.energy.2021.120267.
[6] Wang R, Wen X, Wang X, Fu Y, Zhang Y. Low carbon optimal operation of integrated energy system based on carbon capture technology, LCA carbon emissions and ladder-type carbon trading. Appl Energy 2022;311:118664. https://doi.org/10.1016/j.apenergy.2022.118664.
[7] Zhang L, Li S, Nie Q, Hu Y. A two-stage benefit optimization and multi-participant benefit-sharing strategy for hybrid renewable energy systems in rural areas under carbon trading. Renew Energy 2022;189:744–61. https://doi.org/10.1016/j.renene.2022.03.048.
[8] Feng F, Du X, Si Q, Cai H. Hybrid game optimization of microgrid cluster (MC) based on service provider (SP) and tiered carbon price. Energies 2022;15(14):14. https://doi.org/10.3390/en15145291.
[9] Cheng Z, et al. Multi-stage planning of park-level integrated energy system based on ladder-type carbon trading. Electric Power Automat Equipment 2021;41(09):148–55. https://doi.org/10.16081/j.epae.202109012.
[10] Kang C, Zhou T, Chen Q, Xu Q, Xia Q, Ji Z. Carbon emission flow in networks. Sci Rep 2012;2(1):1. https://doi.org/10.1038/srep00479.
[11] Ministry of Ecology and Environment of the People's Republic of China. The interim rules for carbon emissions trading management [EB/OL]. (2021-01-05) [2022-11-07]. https://www.mee.gov.cn/xxgk2018/xxgk/xxgk02/ 202101/t20210105_816131.html.
[12] Cheng Y, Zhang N, Wang Y, Yang J, Kang C, Xia Q. Modeling carbon emission flow in multiple energy systems. IEEE Trans Smart Grid 2019;10(4):3562–74. https://doi.org/10.1109/TSG.2018.2830775.
[13] Cheng Y, Zhang N, Wang Y, Yang J, Kang C, Xia Q. Modeling carbon emission flow in multiple energy systems. In: 2019 IEEE Power & Energy Society General Meeting (PESGM); 2019. p. 1. https://doi.org/10.1109/PESGM40551.2019.8974086.
[14] J. Cao, X. Wu, Y. Hu, Z. Zhang, P. Wang, and Y. Wu, Low-carbon planning of integrated energy system considering a reward and punishment ladder-type carbon trading mechanism. In: 2022 IEEE 6th Conference on Energy Internet and Energy System Integration (EI2), 2022, pp. 2671–2676. doi: 10.1109/EI256261.2022.10116827.
[15] Li J, He X, Li W, Zhang M, Wu J. Low-carbon optimal learning scheduling of the power system based on carbon capture system and carbon emission flow theory. Electr Pow Syst Res 2023;218:109215. https://doi.org/10.1016/j.epsr.2023.109215.
[16] Liu Z, Xing H, Luo Y, Ye Y, Shi Y. Low-carbon economic dispatch of an integrated energy system based on carbon emission flow theory. J Electr Eng Technol 2023;18(3):1613–24. https://doi.org/10.1007/s42835-022-01298-7.
[17] Shao C, Ding Y, Wang J. A low-carbon economic dispatch model incorporated with consumption-side emission penalty scheme. Appl Energy 2019;238:1084–92. https://doi.org/10.1016/j.apenergy.2019.01.108.
[18] Specht JM, Madlener R. Energy Supplier 2.0: A conceptual business model for energy suppliers aggregating flexible distributed assets and policy issues raised. Energy Policy 2019;135:110911. https://doi.org/10.1016/j.enpol.2019.110911.
[19] Liu P, Ding T, Zou Z, Yang Y. Integrated demand response for a load serving entity in multi-energy market considering network constraints. Appl Energy 2019;250:512–29. https://doi.org/10.1016/j.apenergy.2019.05.003.
[20] Li Z, Wu W, Shahidehpour M, Wang J, Zhang B. Combined heat and power dispatch considering pipeline energy storage of district heating network. IEEE Trans Sustainable Energy 2016;7(1):12–22. https://doi.org/10.1109/TSTE.2015.2467383.
[21] Wu C, Gu W, Jiang P, Li Z, Cai H, Li B. Combined economic dispatch considering the time-delay of district heating network and multi-regional indoor temperature control. IEEE Trans Sustainable Energy 2018;9(1):118–27. https://doi.org/10.1109/TSTE.2017.2718031.
[22] Li Z, Wu W, Wang J, Zhang B, Zheng T. Transmission-constrained unit commitment considering combined electricity and district heating networks. IEEE Trans Sustainable Energy 2016;7(2):480–92. https://doi.org/10.1109/TSTE.2015.2500571.
[23] Dalla Rosa A, Boulter R, Church K, Svendsen S. District heating (DH) network design and operation toward a system-wide methodology for optimizing renewable energy solutions (SMORES) in Canada: A case study. Energy 2012;45(1):960–74. https://doi.org/10.1016/j.energy.2012.06.062.
[24] Xu Y, et al. Combined Heat and Power Dispatch Based on Integrated Demand Response and Heat Transmission for Wind Power Accommodation. In: 2018 IEEE Power Energy Society General Meeting (PESGM); 2018. p. 1–5. https://doi.org/10.1109/PESGM.2018.8586501.
[25] Chen Y, Xu Y, Li Z, Feng X. Optimally coordinated dispatch of combined-heat-and-electrical network with demand response. IET Gener Transmis Distrib 2019;13(11):2216–25. https://doi.org/10.1049/iet-gtd.2018.6992.
[26] Shao C, Ding Y, Siano P, Lin Z. A framework for incorporating demand response of smart buildings into the integrated heat and electricity energy system. IEEE Trans Ind Electron 2019;66(2):1465–75. https://doi.org/10.1109/TIE.2017.2784393.
[27] Bahrami S, Sheikhi A. From demand response in smart grid toward integrated demand response in smart energy hub. IEEE Trans Smart Grid 2016;7(2):650–8. https://doi.org/10.1109/TSG.2015.2464374.
[28] Jiang T, Li Z, Jin X, Chen H, Li X, Mu Y. Flexible operation of active distribution network using integrated smart buildings with heating, ventilation and air-conditioning systems. Appl Energy 2018;226:181–96. https://doi.org/10.1016/j.apenergy.2018.05.091.
[29] Li X, Li W, Zhang R, Jiang T, Chen H, Li G. Collaborative scheduling and flexibility assessment of integrated electricity and district heating systems utilizing thermal inertia of district heating network and aggregated buildings. Appl Energy 2020;258:114021. https://doi.org/10.1016/j.apenergy.2019.114021.
[30] Chen H, Zhang Y, Zhang R, Lin C, Jiang T, Li X. Privacy-preserving distributed optimal scheduling of regional integrated energy system considering different heating modes of buildings. Energ Conver Manage 2021;237:114096.
[31] Ma T, Pei W, Xiao H, Kong L, Mu Y, Pu T. The energy management strategies based on dynamic energy pricing for community integrated energy system considering the interactions between suppliers and users. Energy 2020;211:118677. https://doi.org/10.1016/j.energy.2020.118677.
[32] Wei F, Jing ZX, Wu PZ, Wu QH. A Stackelberg game approach for multiple energies trading in integrated energy systems. Appl Energy 2017;200:315–29. https://doi.org/10.1016/j.apenergy.2017.05.001.
[33] J. Yang, X. Wang, and S. Zhang, Game Analysis of Multi-energy Markets Considering Integrated Energy Supplier Bidding. In: *2021 6th International Conference on Power and Renewable Energy (ICPRE)*, Sep. 2021, pp. 1440–1445. doi: 10.1109/ICPRE52634.2021.9635401.
[34] Cheng Y, Zhang N, Zhang B, Kang C, Xi W, Feng M. Low-carbon operation of multiple energy systems based on energy-carbon integrated prices. IEEE Trans Smart Grid 2020;11(2):1307–18. https://doi.org/10.1109/TSG.2019.2935736.
[35] Zhou Q. The study of carbon emission obligation allocation in power systems. Shanghai Jiao Tong University 2016. https://doi.org/10.27307/d.cnki.gsjtu.2016.000273.
[36] Chen C, Wang J, Qiu F, Zhao D. Resilient distribution system by microgrids formation after natural disasters. IEEE Trans Smart Grid 2015;7(2):958–66.
[37] Bao Z, Ye Y, Liu R, Cheng W, Zhao Q, Wu T. Scheduling coordination of back pressure CHP coupled electricity-heat energy system with adaptive constraint strategy to accommodate uncertain wind power. Energy 2022;240:122791. https://doi.org/10.1016/j.energy.2021.122791.
[38] He P, et al. District Heating Engineering. China: China Architecture and Building Press; 2009.

[39] Zeng G, et al. Analysis on the environment and synthesis factors affecting the PMV index. Building Energy Efficiency 2007;03:11–6.
[40] Lin L, Gu J, Wang Q. Optimal dispatching of combined heat-power system considering characteristics of thermal network and thermal comfort elasticity for wind power accommodation. Power System Technology 2019;43(10):3648–61. https://doi.org/10.13335/j.1000-3673.pst.2018.2915.
[41] Boyd S. Distributed optimization and statistical learning via the alternating direction method of multipliers. FNT Machine Learn 2010;3(1):1–122. https://doi.org/10.1561/2200000016.